

Project Overview

Pin No	23622-CM-005 , 23622-CM-006, 23622-CM-007, 23622-CM-016
Collage	Geethanjali Institute of Science & technology
Team ID	LTVIP2025TMID60720
Project Name	Calculating family Expenses Using ServiceNow
Team Members	Batturu Narendra, Bala Achith Kumar Reddy Chundi, Boggala Omsai, Botta Vishnu
Team Size	4
Team Leader	Batturu Narendra

Introduction

Introduction to Calculating Family Expenses Using ServiceNow

In today's fast-paced world, effective management of household finances is essential to ensure financial security and informed decision-making. Families often struggle to keep track of various expenses such as groceries, utilities, education, healthcare, transportation, and leisure. Without a centralized system, calculating and analyzing these expenses can become overwhelming and inaccurate.

This project focuses on building a solution for calculating and managing family expenses using the ServiceNow platform. Traditionally used for enterprise IT service management, ServiceNow also offers powerful low-code tools, database capabilities, and automated workflows that can be customized to serve non-IT purposes—such as personal financial tracking.

The main aim of this project is to design and implement a ServiceNow-based application that allows users to:

- Record and categorize daily or monthly family expenses
- Automatically calculate total expenses across different categories
- Set budget thresholds and receive notifications when limits are exceeded
- Generate reports and dashboards for better financial insights

By using ServiceNow, this project not only showcases an innovative application of the platform outside its usual IT context but also demonstrates how workflow automation and data visibility can simplify everyday tasks like calculating and managing household expenses.

Project Overview

Project Title:

Calculating and Managing Family Expenses Using ServiceNow

Objective:

The main goal of this project is to develop a streamlined solution using the ServiceNow platform to track, categorize, and analyze family expenses. By leveraging ServiceNow's workflow automation, form design, and reporting capabilities, the project aims to create a centralized and user-friendly system for financial management within a household.

Scope:

- Design custom tables for expense categories such as groceries, utilities, rent, transportation, education, healthcare, etc.
- Create user-friendly forms for inputting daily/weekly/monthly expenses.
- Automate workflows for approvals or notifications (e.g., spending limit alerts).
- Build dashboards and reports to provide visual insights into spending patterns over time.
- Enable role-based access so that different family members can view or add data as appropriate.

Tools and Platform:

- **ServiceNow:** Custom application development using App Engine Studio, Flow Designer, and Reporting.
- **Scripting (optional):** Server-side (GlideRecord) or client-side (JavaScript) scripting for advanced logic.
- **Performance Analytics:** For detailed charts, indicators, and forecasts (if available in the ServiceNow license).

Benefits:

- Real-time visibility into household spending.
- Better budget planning and financial decision-making.
- Reduced manual effort through automated workflows and reporting.
- Data security and access control within the family.

Expected Outcomes:

- A working ServiceNow app tailored for tracking family expenses.
- Dashboards for monthly and yearly expense summaries.
- Alerts or recommendations when budgets are exceeded or spending trends change

IDEATION PHASE

Problem Statement

Managing household expenses is a common yet often inefficient task for families, typically performed using manual methods such as handwritten notes, spreadsheets, or generic mobile apps. These approaches lack centralized data storage, automation, categorization, and real-time insights, making it difficult to monitor spending trends or maintain financial discipline. Additionally, they offer limited collaboration among family members and no mechanisms for tracking against predefined budgets.

This project addresses the need for a structured, automated, and user-friendly solution by developing a custom application within the ServiceNow platform. The solution will enable families to systematically record expenses, categorize them, generate real-time reports, and set alerts for overspending. By leveraging ServiceNow's capabilities—such as form customization, workflow automation, and dashboard reporting—the application aims to streamline household financial management and enhance overall budgeting accuracy and transparency.

Empathy Map Canvas

User Persona:

A family member responsible for managing or tracking household expenses (e.g., parent, guardian, or financially aware young adult).

1. Says

- "I don't know where all the money goes each month."
 - "I wish I had an easy way to see what we're spending on groceries or bills."
 - "It's hard to keep track when everyone is spending from the same account."
 - "Budgeting feels overwhelming."
-

2. Thinks

- "Are we spending more than we earn?"
 - "I hope we don't run out of money before the month ends."
 - "I wish there was a smarter way to monitor expenses without doing it manually."
 - "Is there a way to alert us when we exceed our budget?"
-

3. Sees

- Disorganized receipts or unclear bank statements.
- Multiple family members using cash, cards, or online payments with no central log.
- Overlapping or duplicate expense entries.
- Apps/tools not tailored to family needs.

4. Hears

- “We need to cut down on unnecessary expenses.”
 - “Why are we always short at the end of the month?”
 - “Can you track how much we spent on utilities this month?”
 - “Let’s start managing our money better.”
-

5. Pains

- Lack of transparency in daily or shared spending.
 - Manual entry and tracking is tedious and error-prone.
 - No reminders or real-time updates on overspending.
 - Difficulty collaborating with other family members on financial matters.
-

6. Gains

- Centralized platform for tracking and analyzing family expenses.
- Automated alerts and reminders for better budgeting.
- Categorized reports for clarity and planning.
- Easier decision-making and improved financial awareness among all members



Brainstorming

The brainstorming phase focused on generating and evaluating ideas to design an efficient and user-friendly expense management solution within the ServiceNow platform. The objective was to determine the key features, technical feasibility, user needs, and automation opportunities that could transform household expense tracking into a centralized and intelligent system.



Key Problem Areas Identified

- Lack of a unified platform for all family members to input and view expenses.
 - Difficulty in tracking spending patterns and identifying budget leaks.
 - Manual effort and inconsistency in entering and maintaining expense records.
 - Limited access control in most personal finance apps.
-



Ideas Generated

1. Expense Entry Form

- Easy-to-use form to capture date, category, amount, and description.
- Support for file uploads (receipts).

2. Expense Categories

- Predefined (e.g., groceries, rent, bills) with ability to add custom categories.

3. Automated Workflows

- Notification if monthly budget limit is crossed.
- Automatic categorization of frequent or recurring expenses.

4. Dashboards & Reports

- Pie charts, bar graphs, and monthly summaries for visual analysis.
- Spending trend comparison over months.

5. User Roles & Permissions

- Parents/guardians: full access.
- Children/others: limited or view-only access.

6. Mobile Compatibility

- Responsive layout for accessing from tablets/phones.

7. Integration Possibilities

- Optional: Connect with bank statement imports or APIs (in advanced versions).

8. Gamification (Optional/Future Enhancement)

- Budget goals and savings challenges to motivate disciplined spending.

✂ Technical Ideas within ServiceNow

- Use **App Engine Studio** to build a scoped application.
- Leverage **Flow Designer** for automation (alerts, reminders).
- Utilize **ServiceNow Tables** for structured data handling.
- Employ **ServiceNow Reports/Dashboards** for real-time insights.
- Optional: Use **Script Includes** or **UI Policies** for dynamic behavior.

REQUIREMENTS ANALYSIS

Customer Journey Map

Stage	User Actions	User Emotions	User Pain Points	Opportunities for Improvement
1. Awareness	Realizes need for expense tracking in the family.	Curious, slightly overwhelmed	Not sure what tool to use; too many options	Promote ServiceNow's customization benefits for families.
2. Research	Looks for tools (apps, Excel, etc.); hears about ServiceNow.	Hopeful, confused	Tools too generic or business-focused	Provide simple examples of personal use cases with ServiceNow.
3. Onboarding	Sets up the ServiceNow expense app with categories.	Motivated, cautious	Setup process may seem complex at first	Include user-friendly templates and guided walkthroughs.
4. Daily Use	Enters daily expenses via form or mobile view.	Comfortable, satisfied	Forgetting to log expenses; repetitive input	Add reminders and auto-fill features for repeat entries.
5. Monitoring	Reviews dashboards and expense reports weekly/monthly.	Engaged, insightful	Charts may be too complex; unclear trends	Offer simplified visuals and spending summaries.
6. Alert/Action	Receives alert when nearing or exceeding budget.	Alerted, cautious	May miss or ignore system notifications	Use email/SMS reminders or color-coded dashboards.
7. Family Discussion	Uses reports to explain finances to other family members.	Empowered, confident	Hard to communicate financial info clearly	Provide printable or shareable summary reports.

Key Takeaways:

- Users value **simplicity, automation, and visual insights**.
- Emotional journey moves from confusion → confidence.
- Every stage is a chance to build trust and usability through **clarity, automation, and accessibility**.

Solution Requirements

1. Functional Requirements

These define what the system should do and the specific features to be implemented.

◆ Expense Management

- The system shall allow users to add, edit, and delete expense records.
- The system shall capture key fields: date, amount, category, payment method, description, and attachment (e.g., receipt).
- The system shall support predefined and user-defined expense categories.

◆ Budgeting

- Users shall be able to set monthly or category-specific budget limits.
- The system shall notify users when they approach or exceed their set budget.

◆ User Roles & Access Control

- The system shall support multiple user roles: Admin (full access) and Member (limited access).
- Admins can view, edit, and delete all entries; Members can add/view only.

◆ Dashboard & Reports

- The system shall display summarized expense data via graphs and charts.
- Reports shall be filterable by date range, category, or user.
- Users shall be able to download or export reports (e.g., PDF, Excel).

◆ Alerts & Notifications

- The system shall trigger alerts when expenses exceed thresholds.
- Users shall receive periodic reminders (e.g., weekly summaries or unentered expenses).

◆ **Data Storage & Logs**

- The system shall store historical expense data securely and allow search/filtering.
 - The system shall maintain logs of user actions for accountability.
-

2. Non-Functional Requirements

These define system qualities, constraints, and performance metrics.

◆ **Usability**

- The system should provide a clean, intuitive interface with minimal training required.
- Forms should be mobile-friendly for quick input from any device.

◆ **Performance**

- The system should be capable of handling up to 1,000 expense records per user without performance degradation.
- Dashboard analytics should load in under 3 seconds.

◆ **Security**

- Only authenticated users shall be able to access the system.
- Data must be protected using role-based access and platform-level encryption (ServiceNow standards).

◆ **Availability**

- The system should have 99.5% uptime as it relies on the ServiceNow platform.
- Users should be able to access the application 24/7 across supported devices.

◆ **Scalability**

- The solution should be scalable to support multi-user households with growing data volumes.
- The architecture should allow future integration with external finance APIs or banking feeds.

Data Flow Diagram

◆ Level 0 DFD (Context Diagram)

Purpose: Shows the system as a single process and its interaction with external entities.

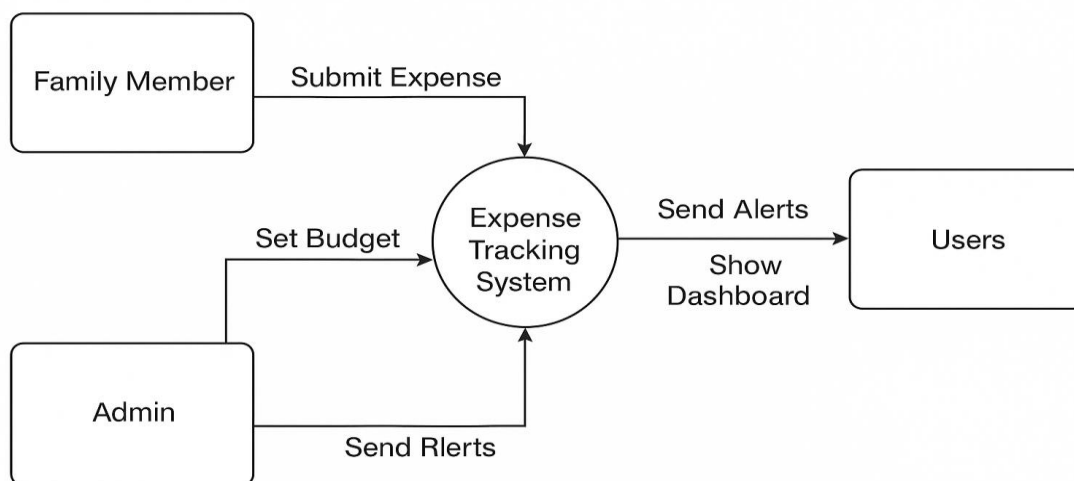
Entities:

1. **Family Member (User)**
2. **Admin User (Primary manager/parent)**
3. **ServiceNow Expense Tracking System (Main Process)**

Flows:

- Family Member → submits expenses → System
- Admin → sets budget, views reports → System
- System → sends alerts/notifications → Admin and Family Member
- System → provides dashboards/reports → Admin and Family Member

Diagram Structure :



◆ Level 1 DFD (Detailed Breakdown)

Purpose: Shows main processes inside the system, internal data stores, and specific flows.

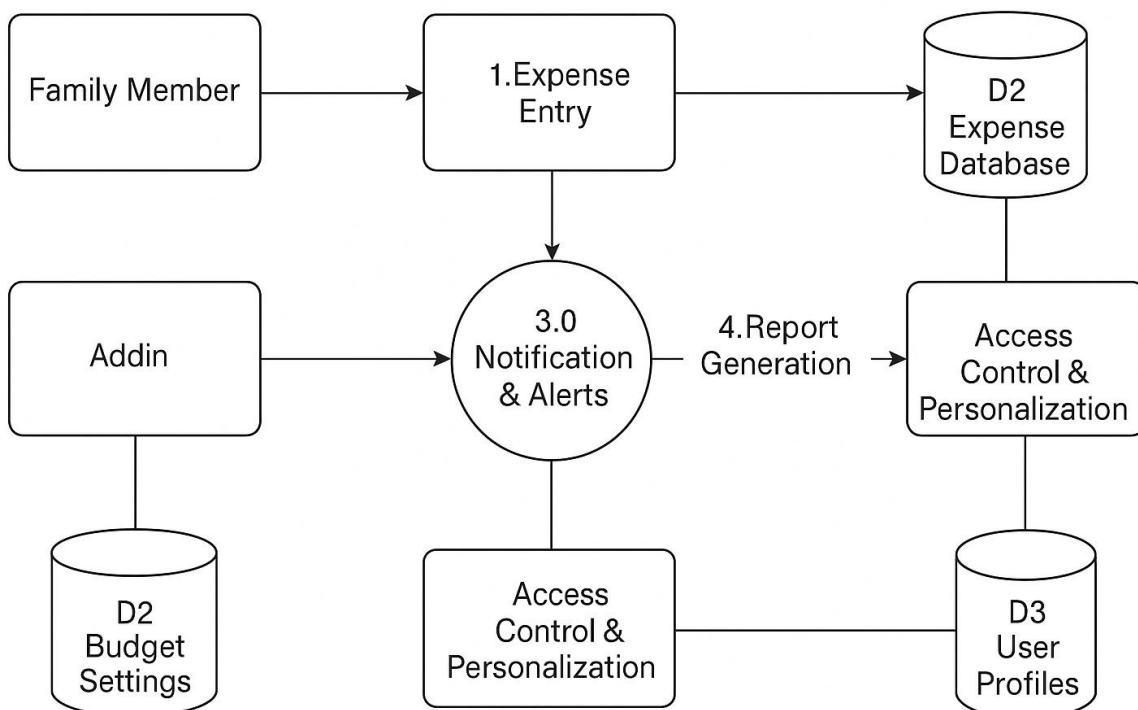
Processes:

- 1.0 Expense Entry
- 2.0 Budget Management
- 3.0 Notification & Alerts
- 4.0 Report Generation

Data Stores:

- D1: Expense Database
- D2: Budget Settings
- D3: User Profiles

Entities & Flows:



✅ Data Stores & Their Purpose:

- **D1: Expense Database** – Stores all submitted expense records
- **D2: Budget Settings** – Stores user-defined budget limits
- **D3: User Profiles** – Stores role-based permissions and user preferences



Technology Stack

◆ Platform

- **ServiceNow**: Core platform for workflow automation, form creation, data storage, and notifications.

◆ Frontend

- **Service Portal** (in ServiceNow): For building user interfaces (Family Member/Admin dashboards).
- **HTML5 / CSS3 / Bootstrap**: Used within the ServiceNow portal for styling and layout.
- **JavaScript**: For client-side scripting and form logic.

◆ Backend

- **ServiceNow Business Rules**: For backend logic like validations, auto-calculations, and triggering alerts.
- **ServiceNow Script Includes / Flow Designer**: For reusable server-side logic and automation flows.
- **GlideRecord API**: For database interactions within ServiceNow.

◆ Database

- **ServiceNow Tables**:
 - u_expense_records
 - u_budget_settings
 - u_user_profiles

◆ Integration (Optional)

- **REST APIs / IntegrationHub:** To connect with external budgeting tools or bank APIs if needed.
- ◆ **Security & Access**
 - **Role-based Access Control (RBAC):** ServiceNow roles for Admin, Family Member, and Viewer.
 - **ServiceNow ACLs (Access Control Lists):** To manage record-level and field-level access.
- ◆ **Reporting & Alerts**
 - **ServiceNow Reports & Dashboards:** For real-time expense tracking.
 - **Notifications Module:** For sending email or in-platform alerts.

PROJECT DESIGN

✓ Problem–Solution Fit

🔍 Identified Problem:

Managing household expenses manually—via spreadsheets, notes, or generic apps—is inefficient and prone to errors. Families often face challenges like:

- Lack of centralized and shared expense tracking.
- Difficulty in monitoring monthly budgets.
- No timely alerts when overspending occurs.
- Limited insights into spending patterns or trends.

💡 Proposed Solution:

Build a **custom ServiceNow application** that offers a centralized, automated platform for tracking, managing, and analyzing family expenses. This solution addresses the key pain points through:

✓ Key Features:

- **Expense Entry Forms:** Simple forms to log daily expenses with categories and descriptions.
- **Budget Management:** Monthly or category-specific limits with warning thresholds.
- **Automated Notifications:** Real-time alerts when expenses exceed set budgets.

- **Dashboards & Reports:** Visual summaries for analyzing spending over time.
- **Role-Based Access:** Secure login with permissions for family members and admin roles.
- **Mobile Compatibility:** Access via any device using ServiceNow's responsive UI.

Why It Fits:

Problem	Solution
No unified system to log expenses	ServiceNow app with shared entry forms and central database
No spending visibility	Dashboards and downloadable reports provide clarity
Overspending without warning	Automated alerts via Flow Designer workflows
Manual and inconsistent tracking	Structured data storage and automation for accuracy and ease
Family coordination around finances is hard	Role-based access and shared dashboards improve transparency and discussion

Fit Confirmation:

By aligning real user pain points with powerful ServiceNow capabilities (automation, workflows, reporting, and security), this project demonstrates a strong **problem–solution fit**. It empowers families to take control of their finances with a structured, scalable tool built on an enterprise-grade platform.

Proposed Solution

Objective:

To develop a custom expense management solution within the ServiceNow platform that enables families to track, analyze, and control their household spending through automated workflows, secure access, and insightful reports.

Solution Overview:

The proposed solution leverages ServiceNow's capabilities to provide a centralized and automated platform for managing family expenses. It simplifies the process of entering,

tracking, and reporting household financial data, while providing alerts and visual summaries to promote financial discipline.

Key Components of the Solution:

1. Expense Entry Module

- Custom form to input expense details (amount, category, date, description).
- Receipt attachment option.
- Mobile-compatible layout for easy input.

2. Budget Management

- Users can define monthly or category-specific budgets.
- Budget thresholds trigger automated alerts via Flow Designer.
- Budgets can be updated as financial needs change.

3. User Roles and Access

- **Admin** (e.g., parent): Full control—manage budgets, view all reports.
- **Member** (e.g., children): Limited access—can log expenses, view personal summaries.
- Access control enforced using ServiceNow ACLs and role permissions.

4. Dashboards & Report

- Real-time visual dashboards showing:
 - Total spending
 - Category breakdown
 - Month-over-month trends
- Exportable reports in Excel or PDF format for family review.

5. Notifications & Alerts

- Alerts for:
 - Exceeding budget
 - Missed expense entries
 - Weekly or monthly summaries

- Notifications sent via ServiceNow's notification engine (email/SMS/portal).

6. Data Storage & Logging

- Expense data stored securely in custom ServiceNow tables.
- Audit trail of user actions for transparency.



Optional Enhancements (Future Scope):

- Bank statement imports via ServiceNow IntegrationHub.
- AI-powered insights for spending recommendations.
- Gamification features for family savings challenges.

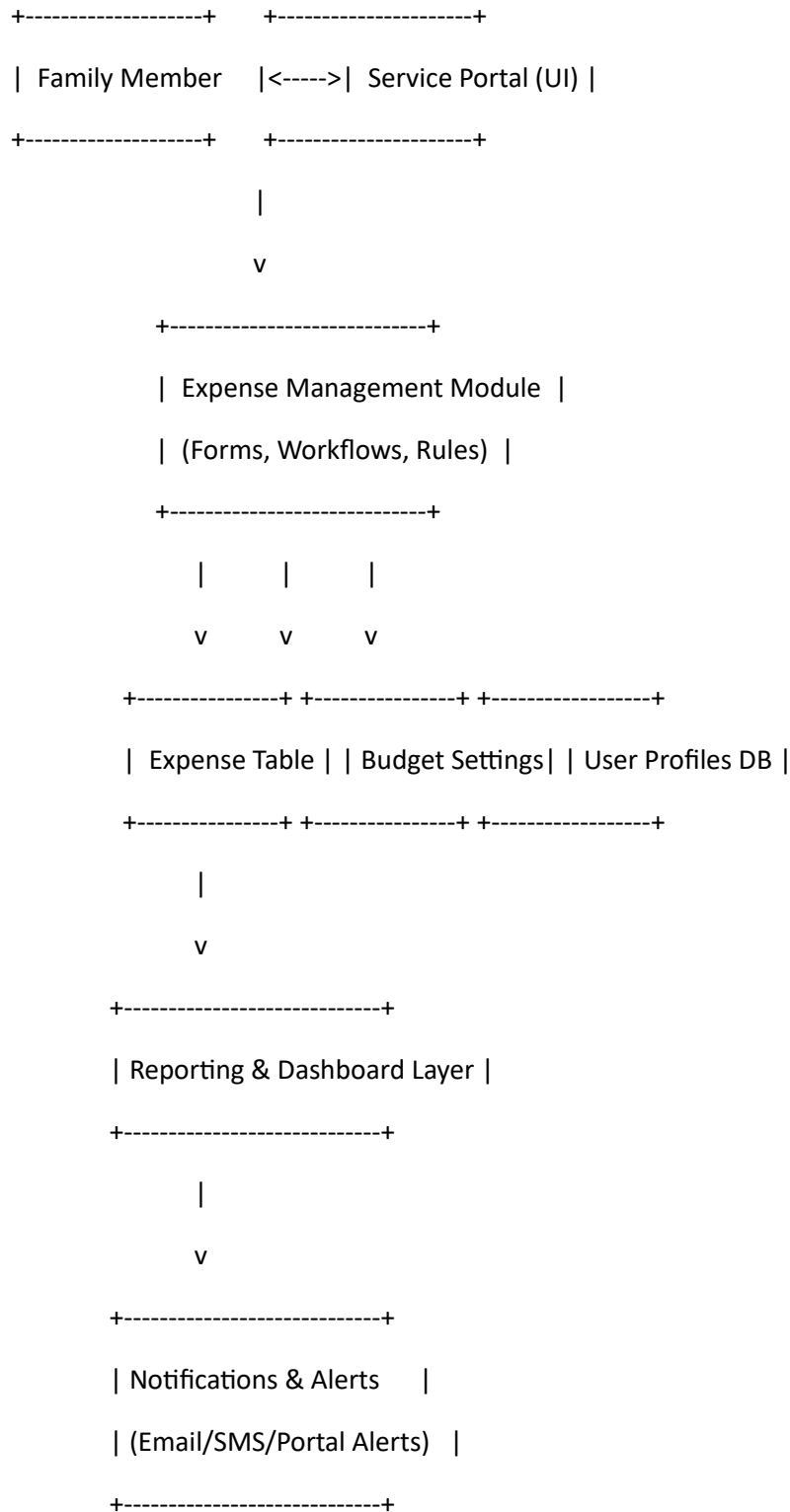


Benefits:

- Centralized and secure expense tracking.
- Improved financial awareness for all family members.
- Automation reduces manual effort and human error.
- Real-time visibility through visual dashboards and reports.
- Scalable for any household size and adaptable to future financial goals.

Solution Architecture

1. High-Level Architecture Overview



2. Core Components and Description

Component	Description
Service Portal (UI)	Web interface for users to submit expenses and view reports. Mobile-friendly.
Expense Form	Custom form for entering and managing expenses, built in ServiceNow Studio.
Budget Management	Interface for admins to define monthly or category budgets.
Flow Designer	Automates alerts when budget limits are crossed.
Business Rules	Validates input, auto-fills categories, handles backend logic.
Tables (CMDB-based)	Custom tables: u_expense_records, u_budget_settings, u_user_profiles.
Dashboard & Reporting	Visual charts for total spend, category spend, trends, etc.
Notification Engine	Sends alerts via email/SMS/portal for overspending or missed logs.
Access Control (RBAC)	Enforced through ServiceNow roles and ACLs (Admin vs Family Member).

3. Security & Access

- **Authentication:** Native ServiceNow user authentication.
- **Authorization:** Role-based access (Admin, Member).
- **Data Protection:** Field-level access via ACLs, platform encryption, audit logs.

4. Integration Points (Optional/Future Scope)

Integration	Purpose
Bank APIs (via IntegrationHub)	Automatically import and categorize expenses.
Google Sheets or Excel	Export monthly reports.
WhatsApp/Email Bot	Log expenses via messaging interfaces.

5. Scalability & Maintainability

- Scales with number of family members or expense entries.
- Easy to extend with additional categories, reports, or AI tools.
- Low maintenance via reusable components and out-of-box ServiceNow workflows.

PROJECT PLANNING & SHEDULING

Project Planning

Project Objective

To design and develop a custom ServiceNow-based application that helps families manage and monitor their monthly expenses with automation, reporting, and alerting features.

Project Phases & Timeline

Phase	Activities	Timeline
1. Requirement Gathering	- Identify user roles (Admin, Family Member) - Define key features and use cases	1 hour
2. Ideation & Design	- Empathy map, brainstorming - Design UI forms and dashboards - Data model planning	1 hour
3. Development – Phase 1	- Build expense form - Configure tables (u_expense_records, etc.) - Implement basic role access	1 hour
4. Development – Phase 2	- Budget setting module - Flow Designer for alerts - Report and dashboard creation	1 hour
5. Testing & QA	- Functional testing - Role-based access testing - Validate reports/notifications	1 hour
6. Deployment & Training	- Deploy on ServiceNow instance - Provide user guide/training for Admin and Family Members	1 hour
7. Feedback & Optimization	- Collect feedback - Fix bugs - Implement small enhancements	1 hour

Team Roles & Responsibilities

Role	Responsibility
Project Manager	Oversee planning, timelines, and team coordination
ServiceNow Developer	Build forms, workflows, tables, and logic in ServiceNow
UX Designer	Design user-friendly forms and dashboards
QA Tester	Perform end-to-end testing and verify alerts/report accuracy
Trainer/User Support	Help onboard family users and document the app usage

Milestones

1.  **Setting up ServiceNow Instance**
2.  **Creation of New Update Set**
3.  **Creation of Table**
4.  **Creation of Table(Daily Expenses)**
5.  **Creation of Relationship**
6.  **Configuring Related List on Family Expenses**
7.  **Creation of Business Rules**
8.  **Configure the Relationship**
9.  **Conclusion**

Risks & Mitigations

Risk	Mitigation Strategy
Complexity in customizing workflows	Use ServiceNow Flow Designer and scripting best practices
User confusion with role access	Provide simple UI and user training
Budget tracking logic may be inaccurate	Conduct thorough test cases with sample data

FUNCTIONAL & PERFORMANCE TESTING

Performance Testing Plan

Project: *Calculating Family Expenses Using ServiceNow*

Objective

To validate that the custom ServiceNow-based family expense tracking system performs reliably and responds within acceptable time limits during typical and peak user activity.

Performance Testing Goals

- Ensure **form submissions** and **dashboard rendering** occur without significant delays.
- Confirm the system can **handle multiple concurrent users** (family members and admins).
- Verify that **automated workflows** (notifications, budget alerts) trigger within the expected time.
- Evaluate the **load capacity** and **response time** of key modules like:
 - Expense Entry
 - Budget Management
 - Report Generation

Test Environment Setup

Component Configuration

Platform	ServiceNow Personal Developer Instance (PDI) or enterprise sandbox
Test Users	5–15 simulated users (admin + members)
Tools	ServiceNow ATF (Automated Test Framework), Postman (API load), or JMeter (optional for APIs)
Test Data	1,000+ dummy expense entries across categories and dates

Test Scenarios

Test Case ID	Scenario	Expected Outcome
PT001	Submit a new expense	Data saved in ≤2s
PT002	Load the budget management screen	Page renders within 3s
PT003	Generate monthly report with 500+ records	Report completes in ≤6s
PT004	10 users logging expenses simultaneously	No crashes or excessive lag
PT005	Budget exceeded – alert trigger	Notification sent within ≤3s
PT006	Dashboard visual update after entry	Update appears in ≤5s

Post-Test Cleanup

- Remove test entries and restore original configurations.
- Reset user sessions and permissions if altered during testing.

Conclusion & Reporting

- Compile results into a performance summary.
- Highlight any bottlenecks or failures with recommendations.
- Re-test after optimizations if needed.

Project Result Phase

Calculating Family Expenses using Service Now

The project aims to develop a comprehensive expense calculation system using ServiceNow. This system will enable users to track and manage family expenses efficiently. It will include features such as expense categorization, budget setting, real-time tracking, and reporting capabilities. Utilizing ServiceNow's robust platform, the project will ensure seamless integration, user-friendly interface, and scalability to accommodate varying family sizes and financial complexities. The end goal is to empower users with the tools they need to make informed financial decisions and promote financial well-being within the family unit.

Milestone-1: **Setting up ServiceNow Instance**

Milestone-2: **Creation of New Update Set**

Milestone-3: **Creation of Table(Daily Expenses)**

Milestone-4: **Creation of Relationship**

Milestone-5: **Configuring Related List on Family Expenses**

Milestone-6: **Creation of Business Rules**

Milestone-7: **Configure the Relationship**

Milestone-8: **Conclusion**

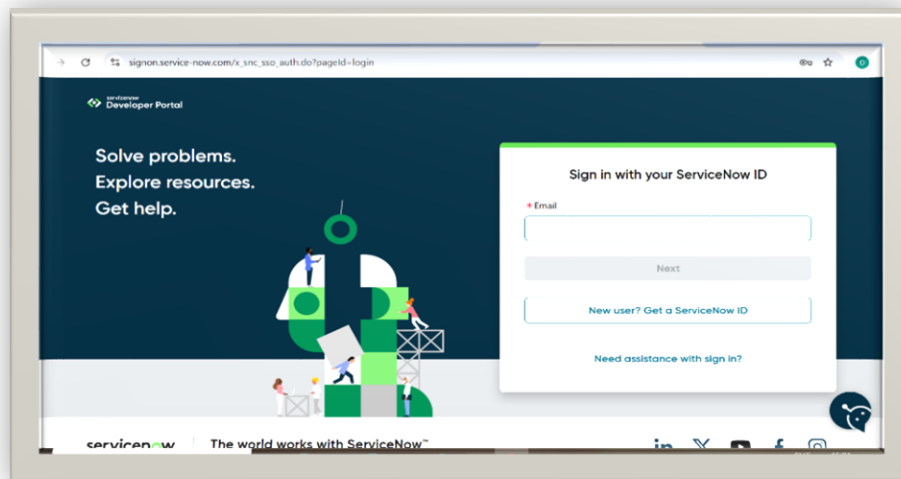
MILESTONE-1

Setting up ServiceNow Instance

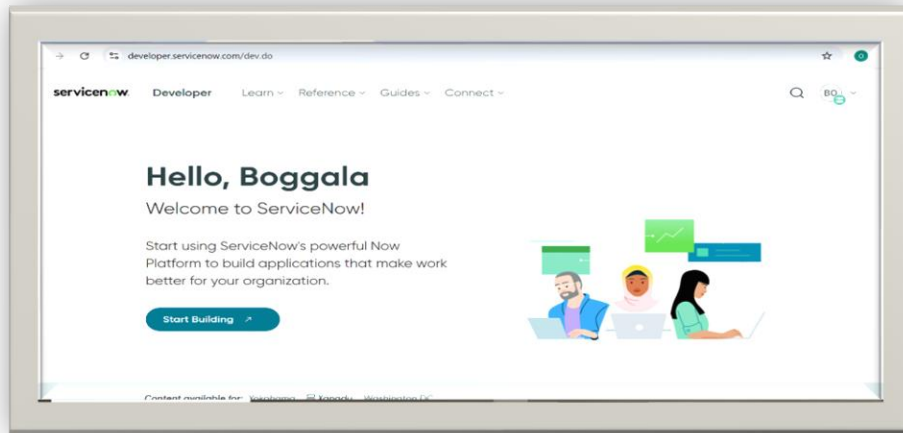
1. Sign up for a developer account on the ServiceNow Developer site
"https://developer.servicenow.com".



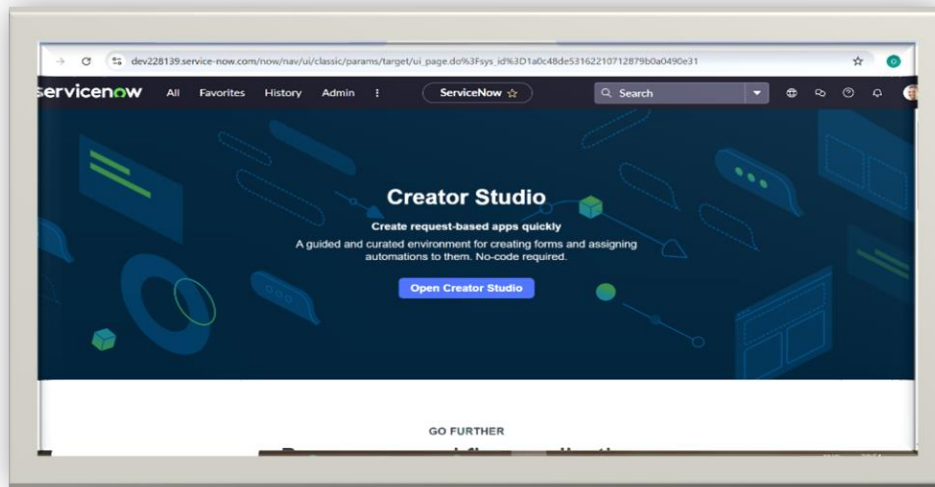
2. Once logged in, navigate to the "Personal Developer Instance" section.
3. Click on "Request Instance" to create a new ServiceNow instance.
4. Fill out the required information and submit the request.
5. You'll receive an email with the instance details once it's ready.
6. Log in to your ServiceNow instance using the provided credentials.



7. Now you will navigate to the ServiceNow.



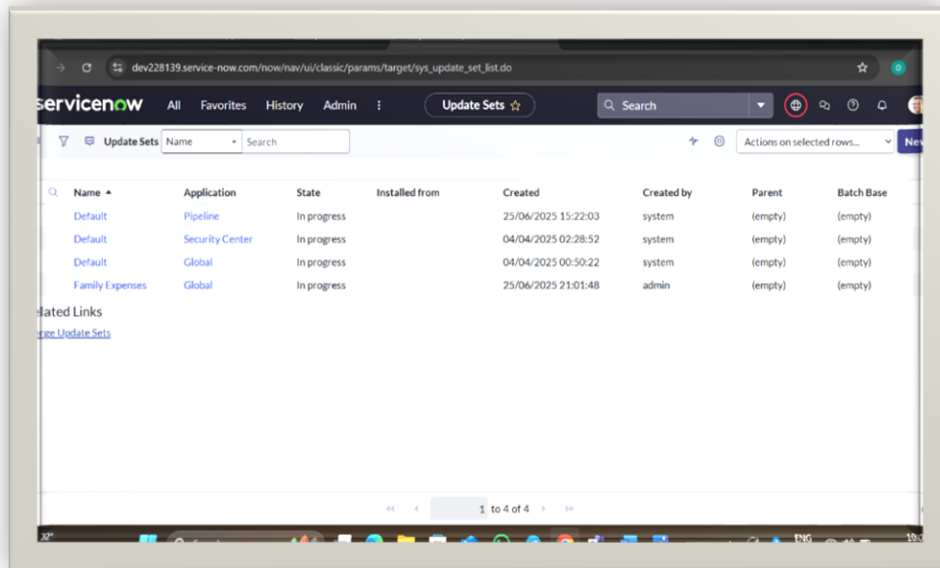
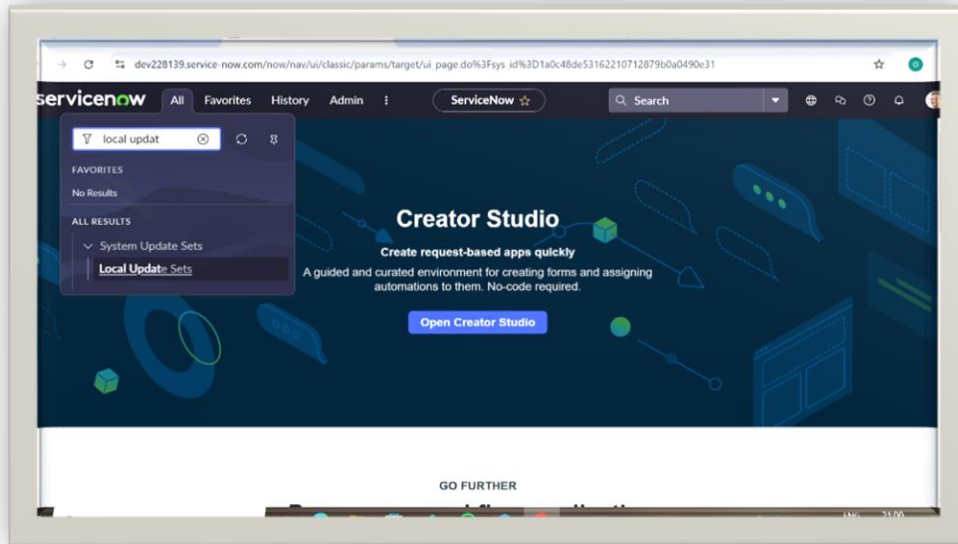
8. Now your instance page is like shown below



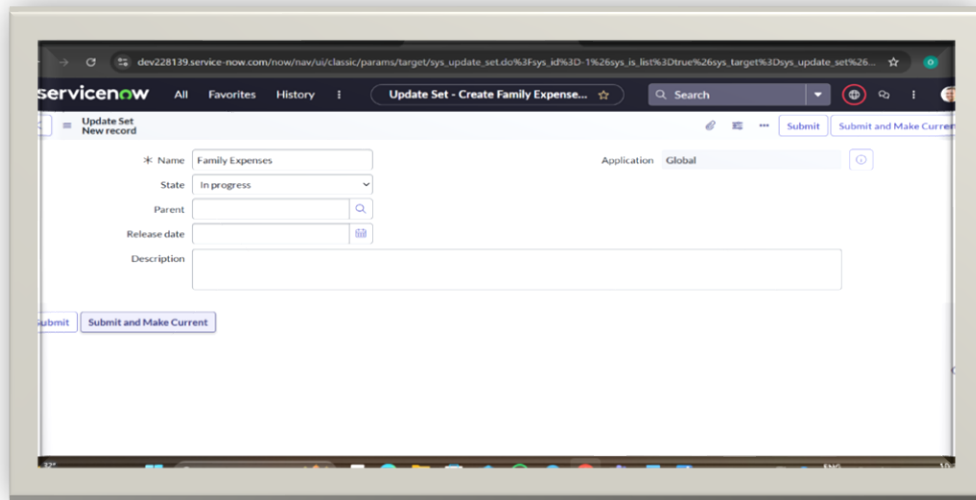
MILESTONE-2

Creation of New Update Set

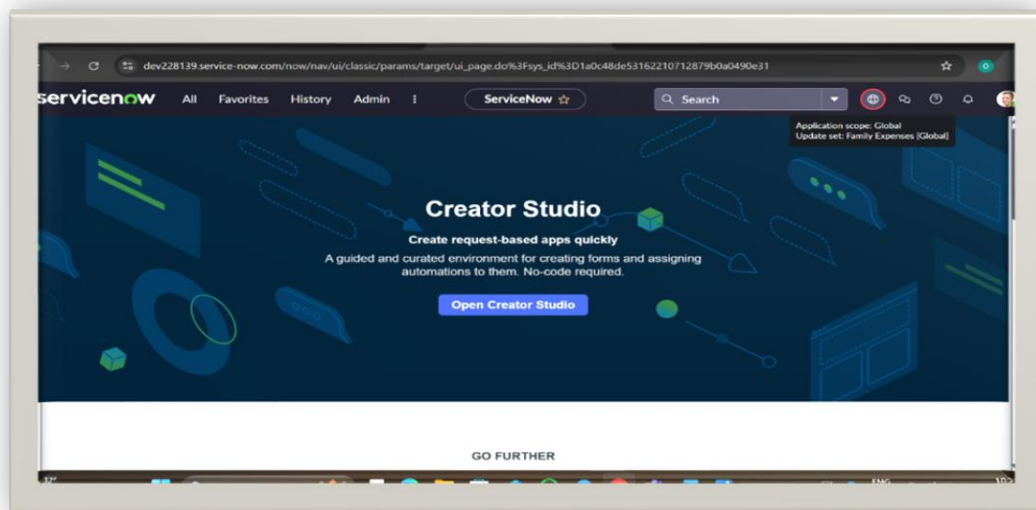
1. Go to All >> In the filter search for Local Update set > click on New.



2. Enter the Details as:
Name : Family Expenses
3. Then click on Submit and Make current.



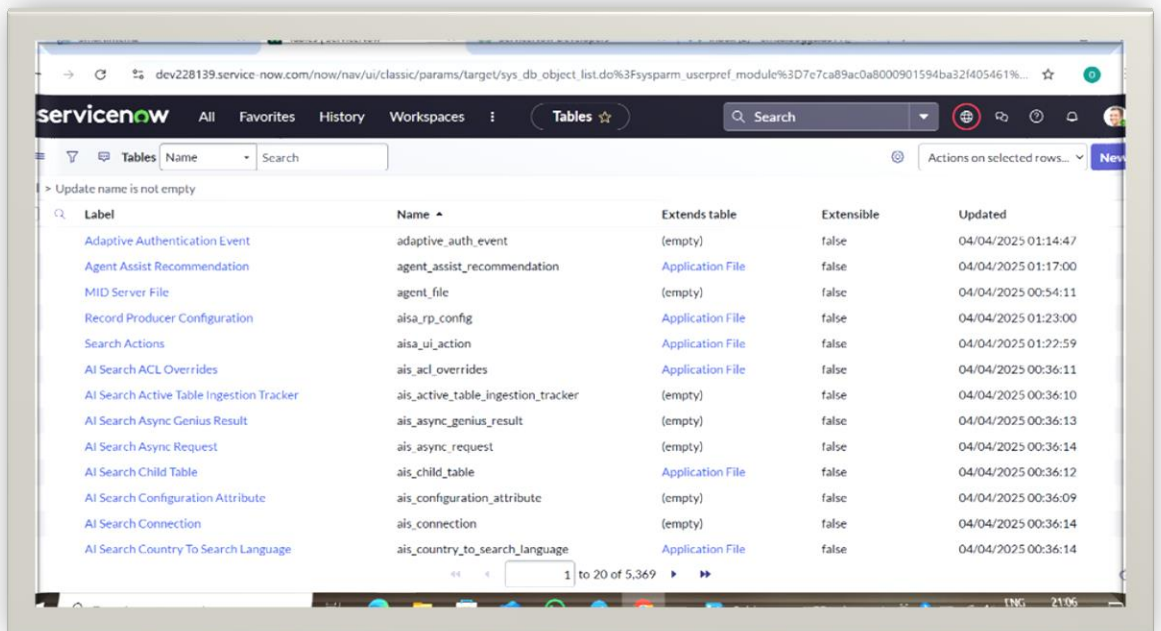
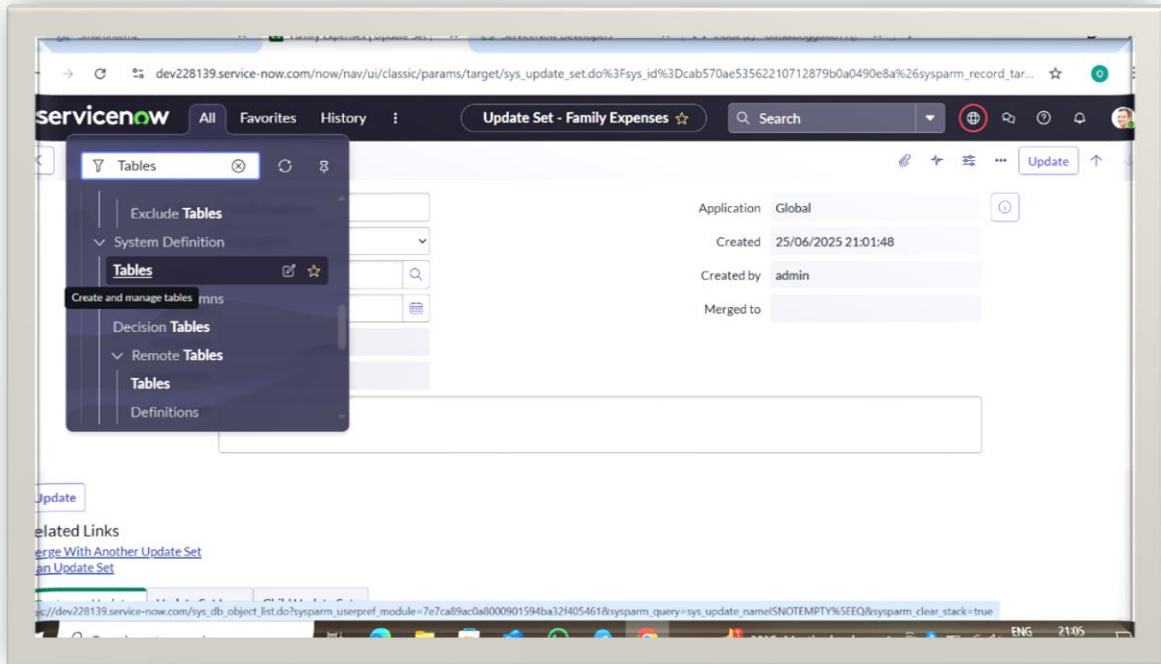
4. After Clicking the submit and Make Current your instance show like this



MILESTONE-3:

ACTIVITY-1: Creation of Family Expenses Table

1. Go to All > In the filter search for Tables > click on New.



2. Enter the Details:

Label : Family Expenses

Name : Auto-Populated

New menu name : Family Expenditure

dev228139.service-now.com/now/nav/ui/classic/params/target/sys_db_object.do%3Fsys_id%3D-1%26sys_is_list%3Dtrue%26sys_target%3Dsys_db_object%26sys...

servicenow All Favorites History Admin Table - New Record Search

Table - New record Submit Cancel

ServiceNow recommends creating custom tables in scoped applications. To learn more about creating scoped applications, click [here](#).

Table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Applications use tables and records to manage data and processes. [More info](#)

* Label Family Expenses Application Global

* Name u_family_expenses Create module ☒

Extends table Create mobile module ☒

Add module to menu -- Create new --

New menu name Family Expenditure

Columns Controls Application Access

Table Columns for text Search

Dictionary Entries

3. Go to the Header and right click there>> click on Save.

dev228139.service-now.com/now/nav/ui/classic/params/target/sys_db_object.do%3Fsys_id%3D-1%26sys_is_list%3Dtrue%26sys_target%3Dsys_db_object%26sys...

servicenow All Favorites History Admin Table - New Record Search

Table - New record Submit Cancel

ServiceNow recommends creating custom tables in scoped applications. To learn more about creating scoped applications, click [here](#).

Table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Applications use tables and records to manage data and processes. [More info](#)

* Label Family Expenses Application Global

* Name u_family_expenses Create module ☒

Extends table Create mobile module ☒

Add module to menu -- Create new --

New menu name Family Expenditure

Columns Controls Application Access

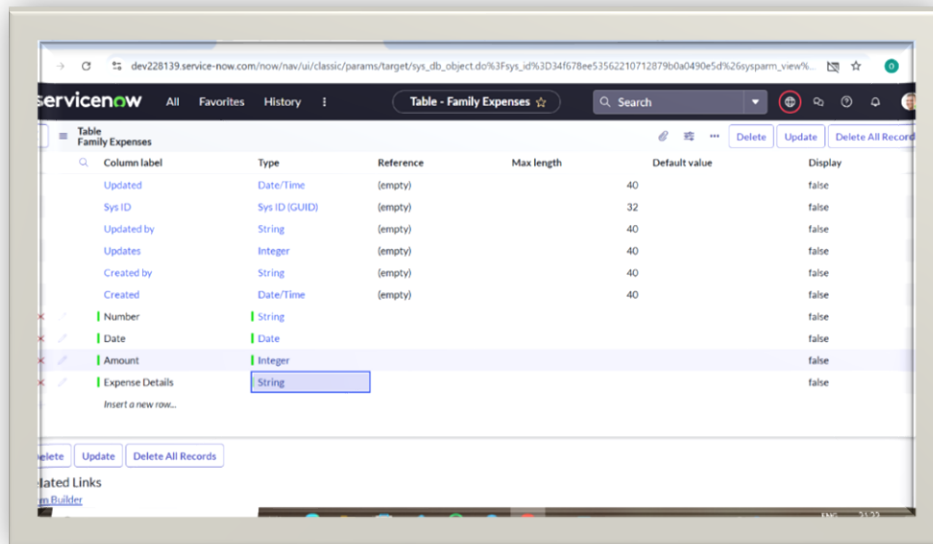
Table Columns for text Search

Dictionary Entries

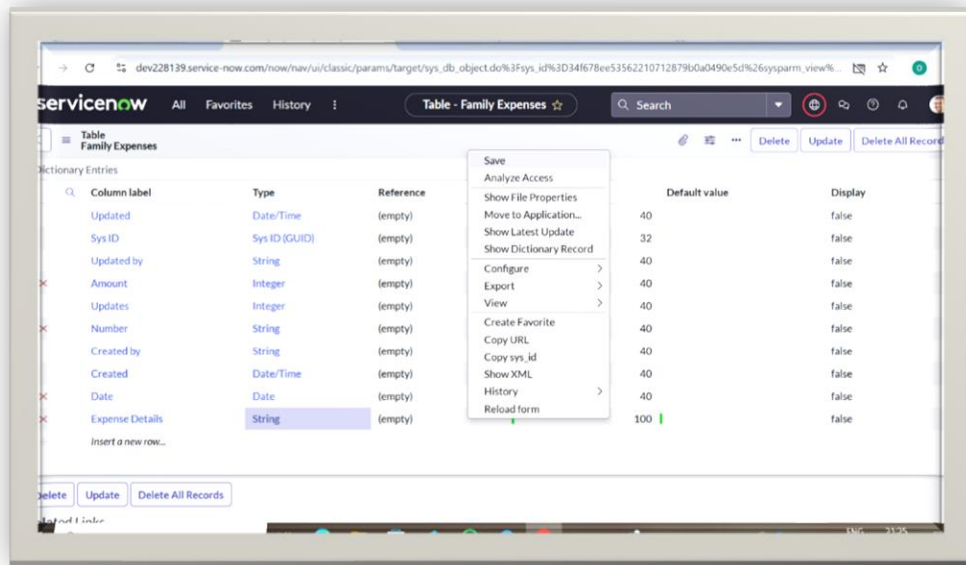
Save
Track in Update Sets
Configure
Export
Create Favorite
Copy URL
Copy sys_id
Reload form

ACTIVITY-2: Creation of Columns (Fields)

1. Near Columns Double click near insert a new row.
2. Give the details as:
Column label: Number
Type: String
3. Double click on insert a new row again
4. Give the details as:
Column label: Date
Type: Date
5. Double click on insert a new row again
6. Give the details as:
Column label: Amount
Type: Integer
7. Double click on insert a new row again
8. Give the details as:
Column label: Expense Details
Type: String
Max length: 800

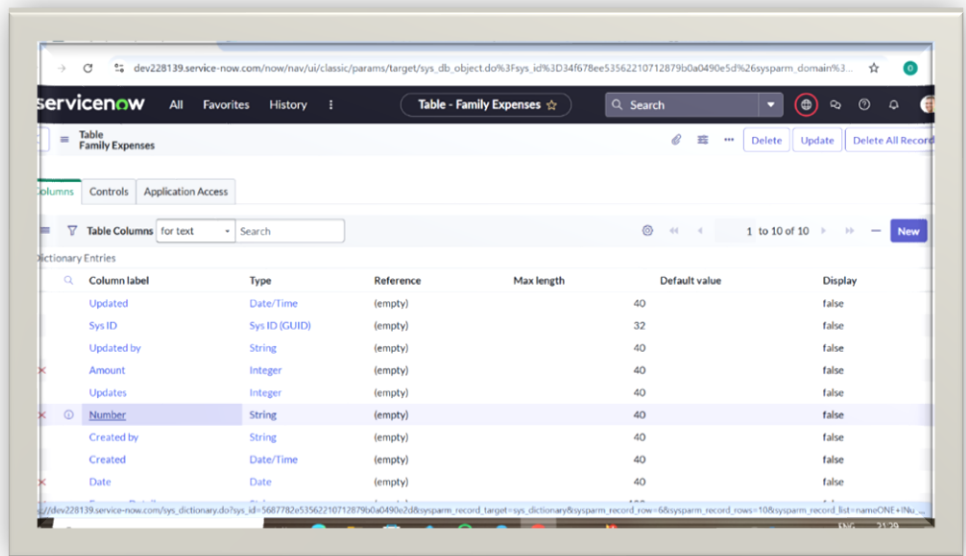


9. Go to the Header and right click there>> click on Save

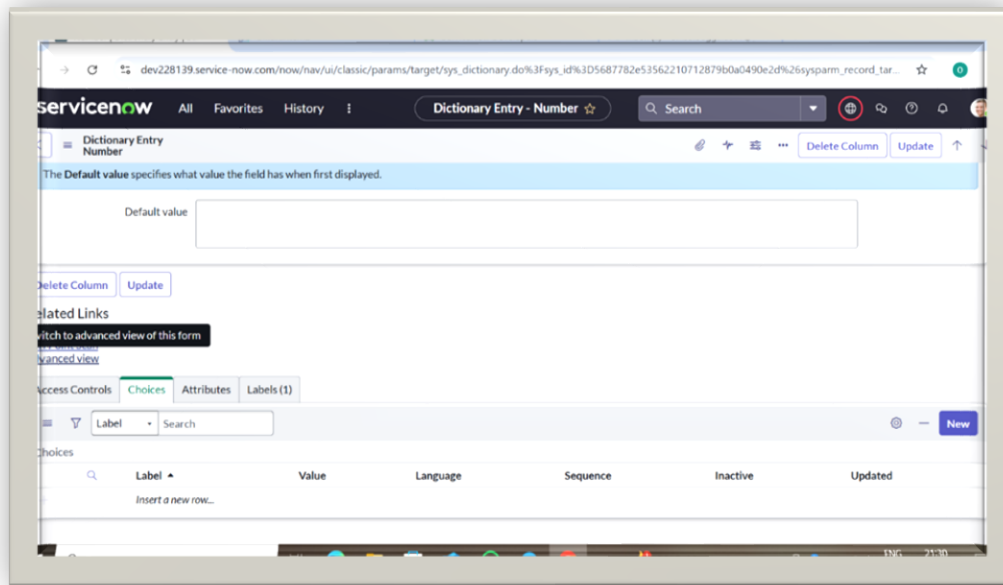


ACTIVITY-3: Making Number Field an Auto-Number

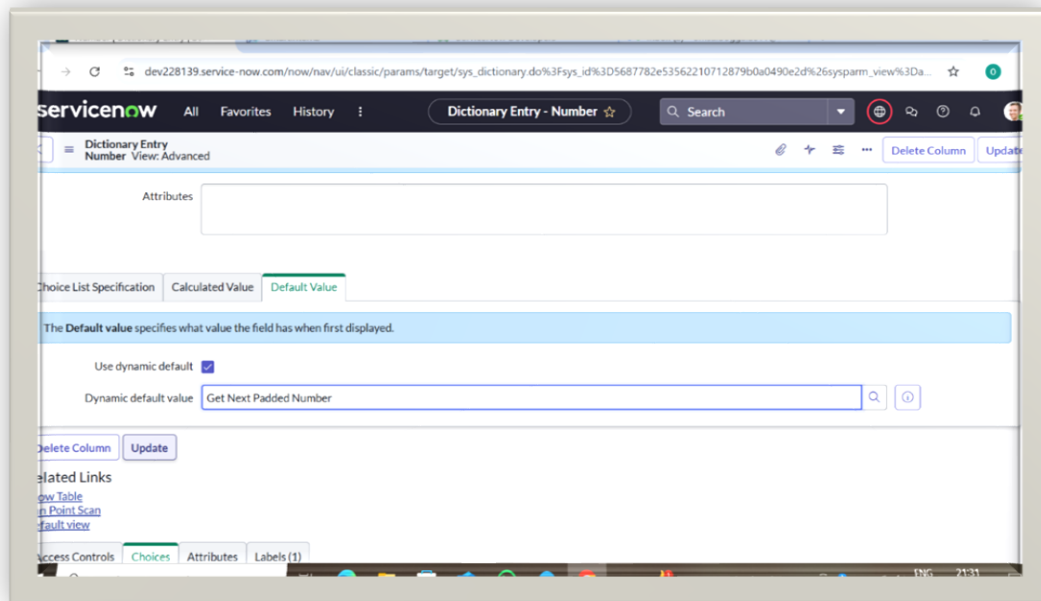
1. Double click on the Number Field/Column.



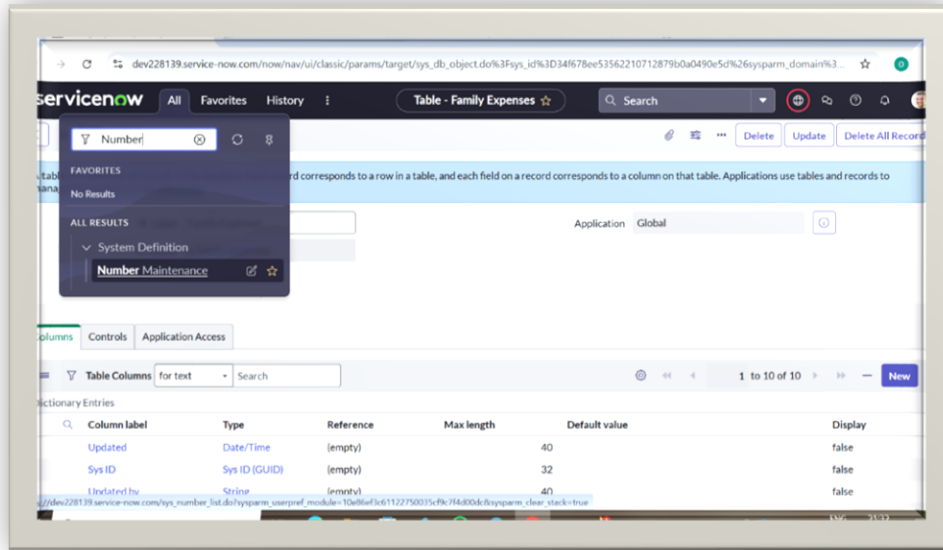
2. Go down and double click on Advanced view



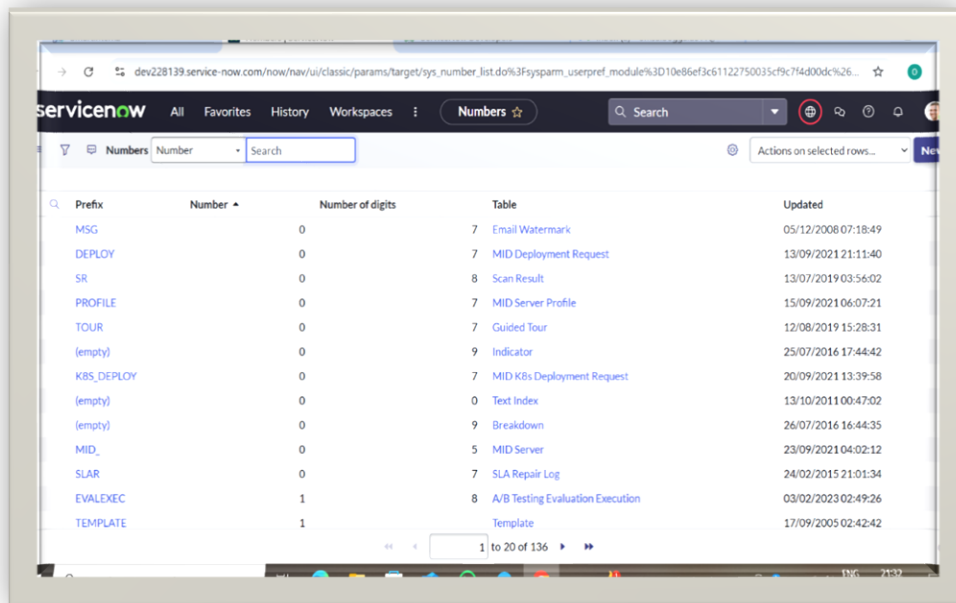
3. In Default Value:
Use dynamic default: check the box
Dynamic default value: Get Next Padded Number
4. Click on Update.



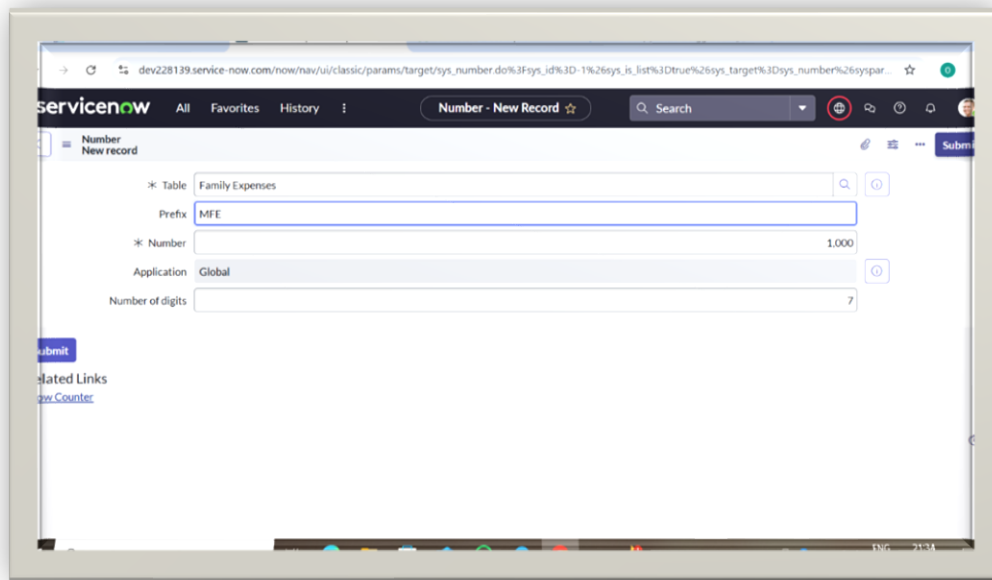
- Go to All >> In the filter search for Number Maintenance >> select Number Maintenance



- Click on New.



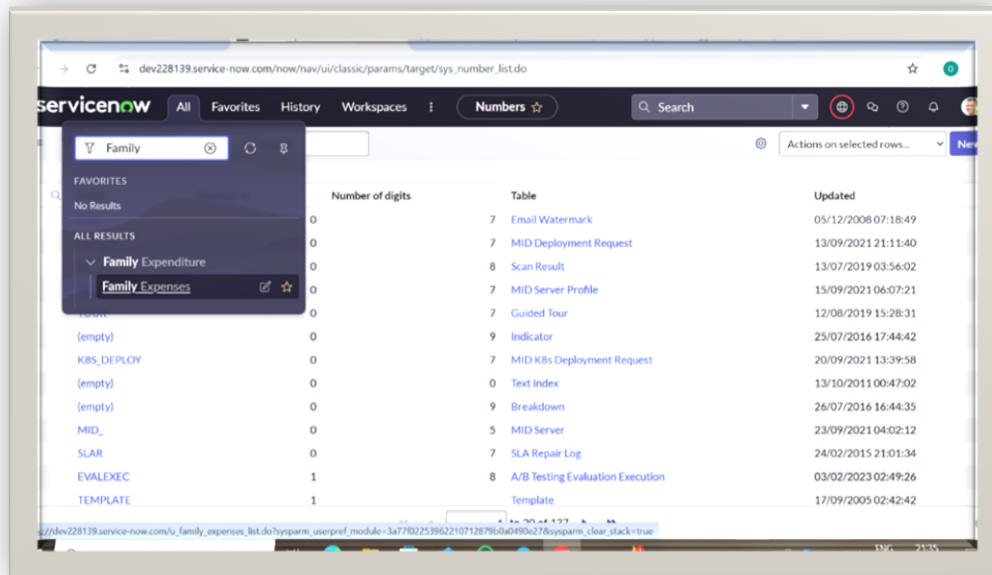
- Enter the below Details:
Table : Family Expenses
Prefix : MFE



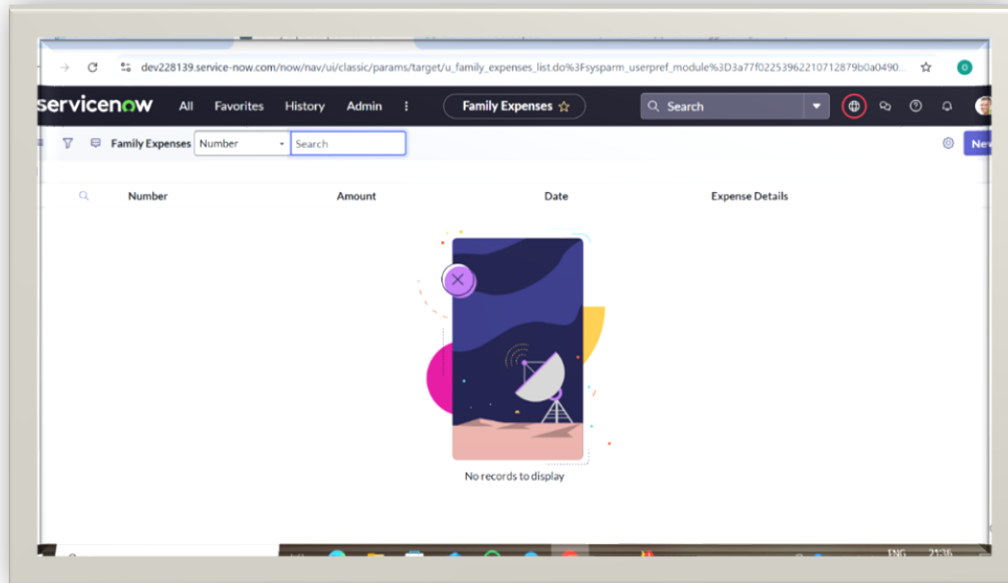
8. Click on Submit.

ACTIVITY-4: Configure the Form

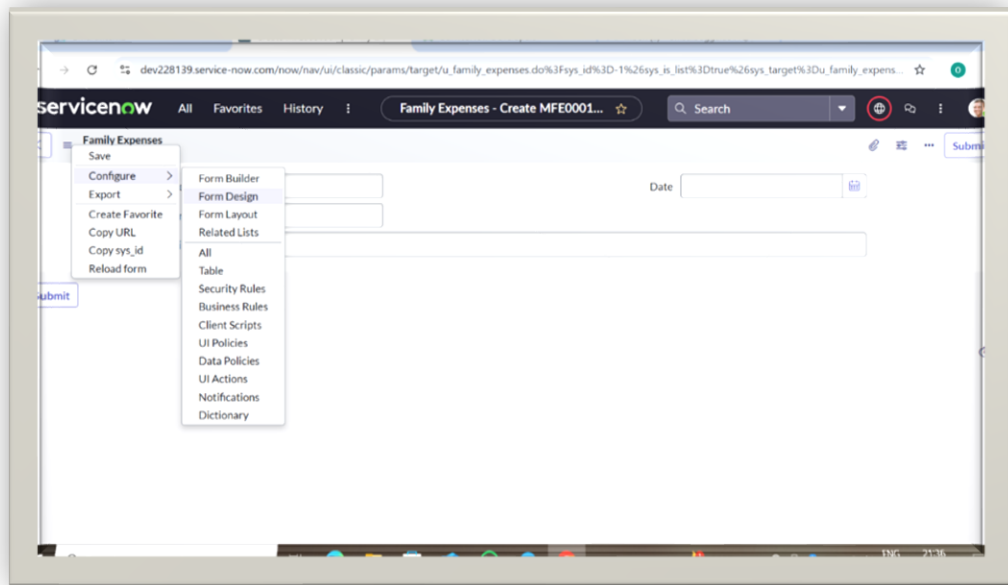
1. Go to All >> In the filter search for Family Expenses >> Open Family Expenses



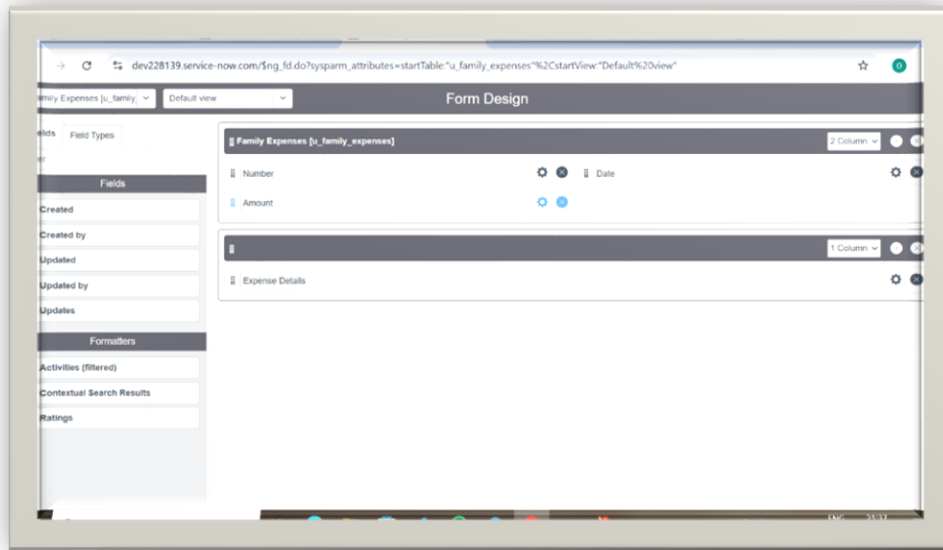
2. Click on New



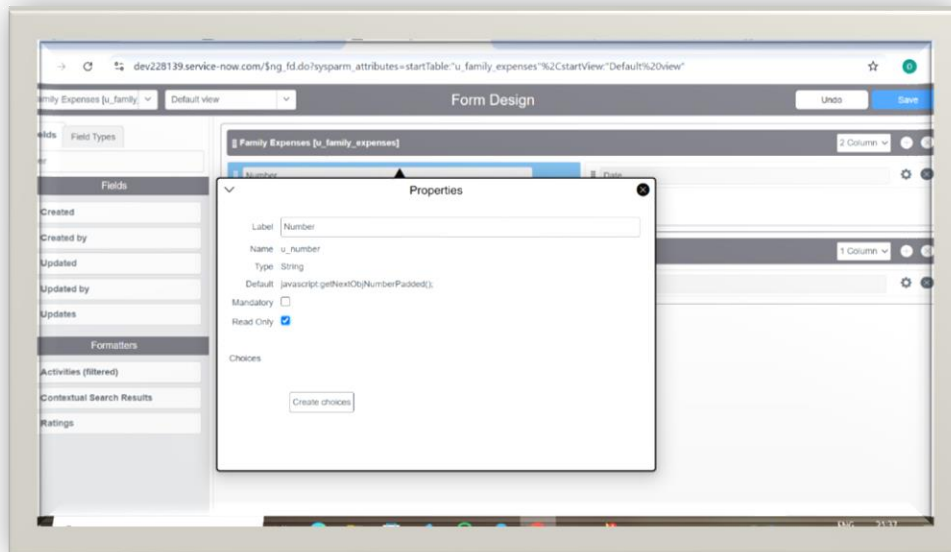
3. Go to the Header and right click there>> click on Configure >> Select Form Design



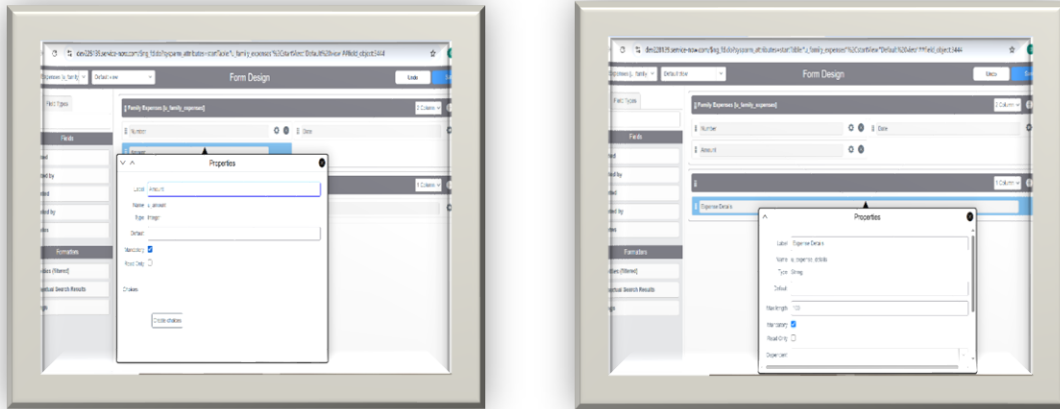
4. Customize or Drag Drop the form as per your requirements



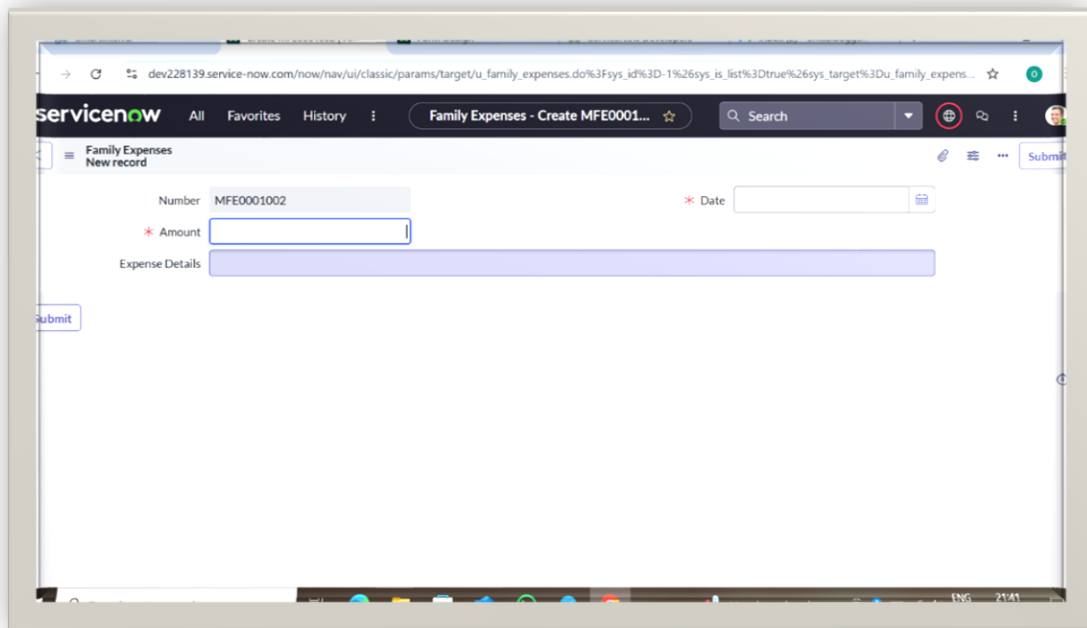
5. Make Number Read-Only Field by clicking on the gear icon and checking Read-Only



6. Make Date, Amount Mandatory Field by clicking on the gear icon and checking Mandatory



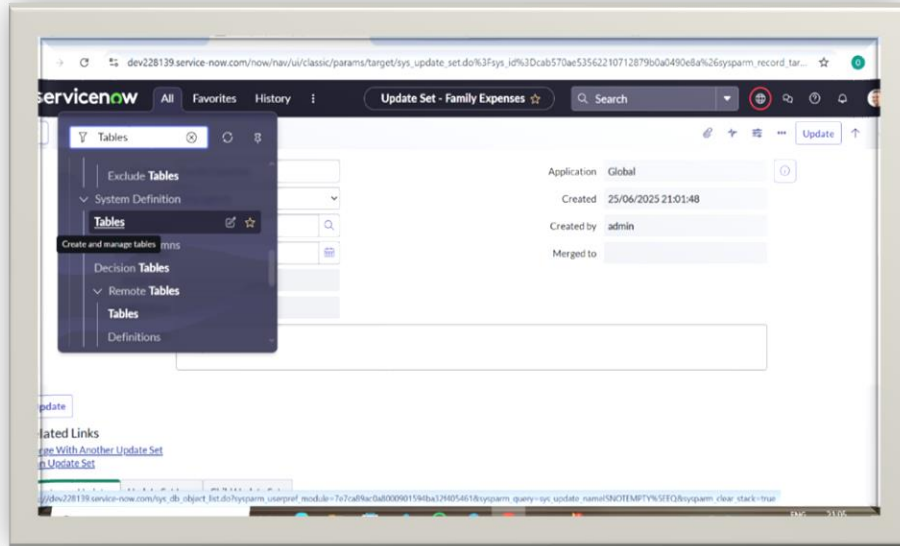
7. Click on Save.



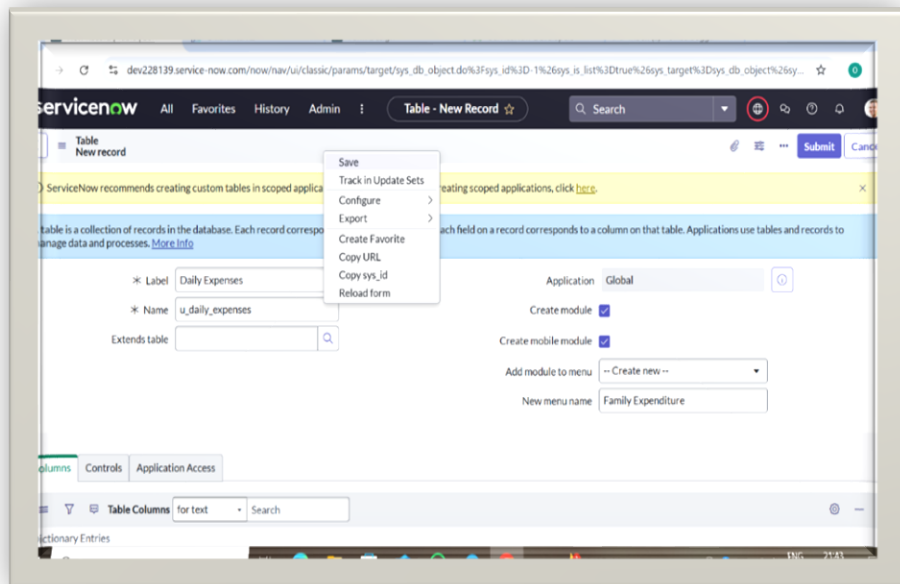
MILESTONE-4:

ACTIVITY-1: Creation of Table (Daily Expenses)

1. Go to All > In the filter search for Tables > click on New.

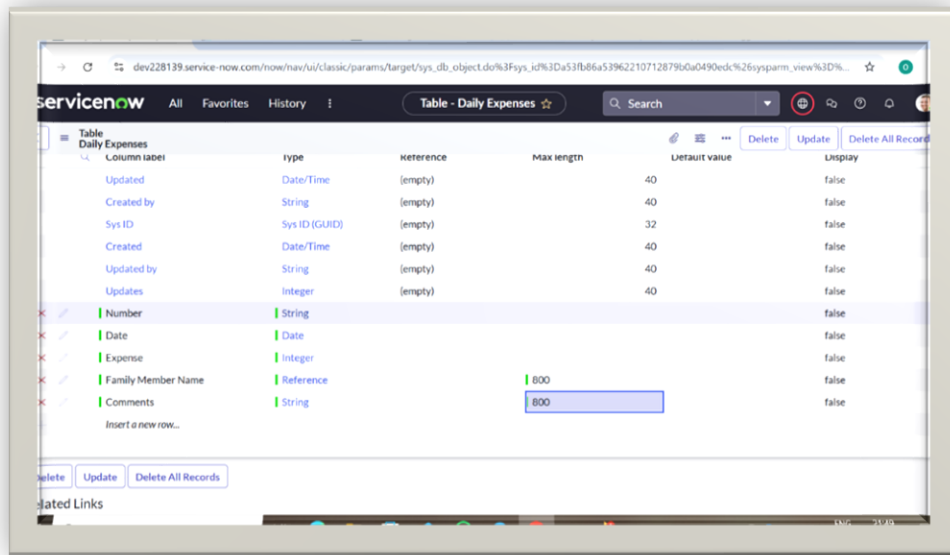


2. Enter the Details:
Label: Daily Expenses
Name: Auto-Populated
Add Module to menu: Family Expenditure
3. Go to the Header and right click there>> click on Save.



ACTIVITY-2: Creation of Columns (Fields)

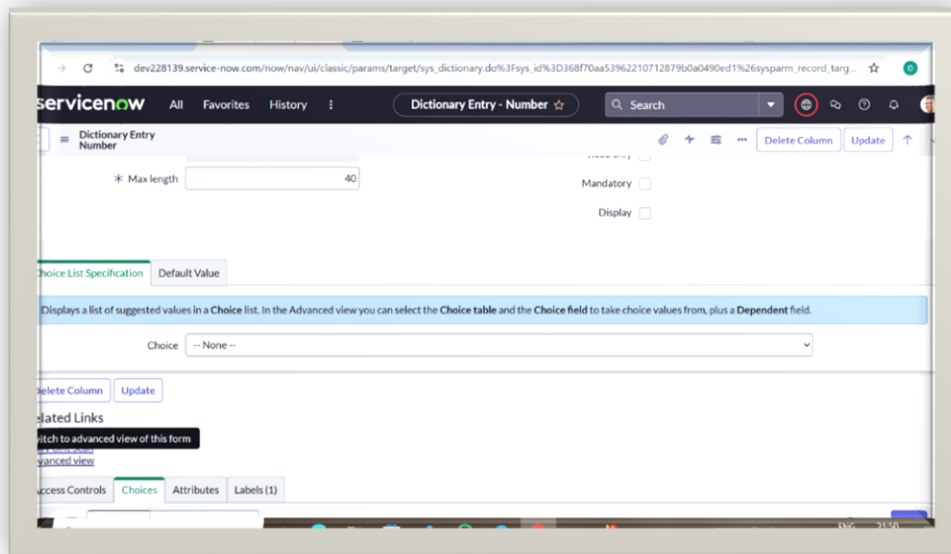
1. Near Columns Double click near insert a new row.
2. Give the details as:
 Column label: Number
 Type: String
3. Double click on insert a new row again
4. Give the details as:
 Column label: Date
 Type: Date
5. Double click on insert a new row again
6. Give the details as:
 Column label: Expense
 Type: Integer
7. Double click on insert a new row again
8. Give the details as:
 Column label: Family Member Name
 Type: Reference
 Max length: 800
9. Double click on insert a new row again
10. Give the details as:
 Column label: Comments
 Type: String
 Max length: 800



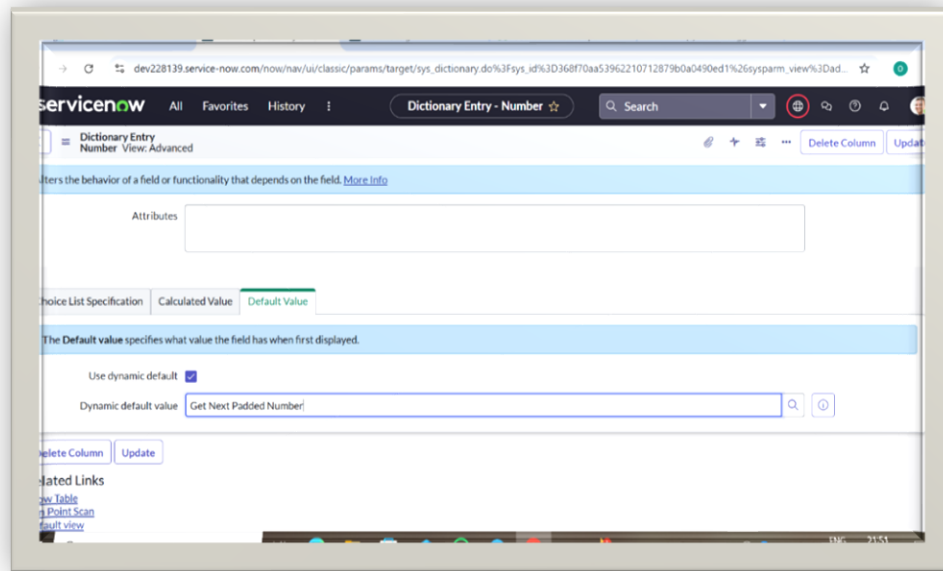
10. Go to the Header and right click there>> click on Save.

ACTIVITY-3: Making Number Field an Auto-Number

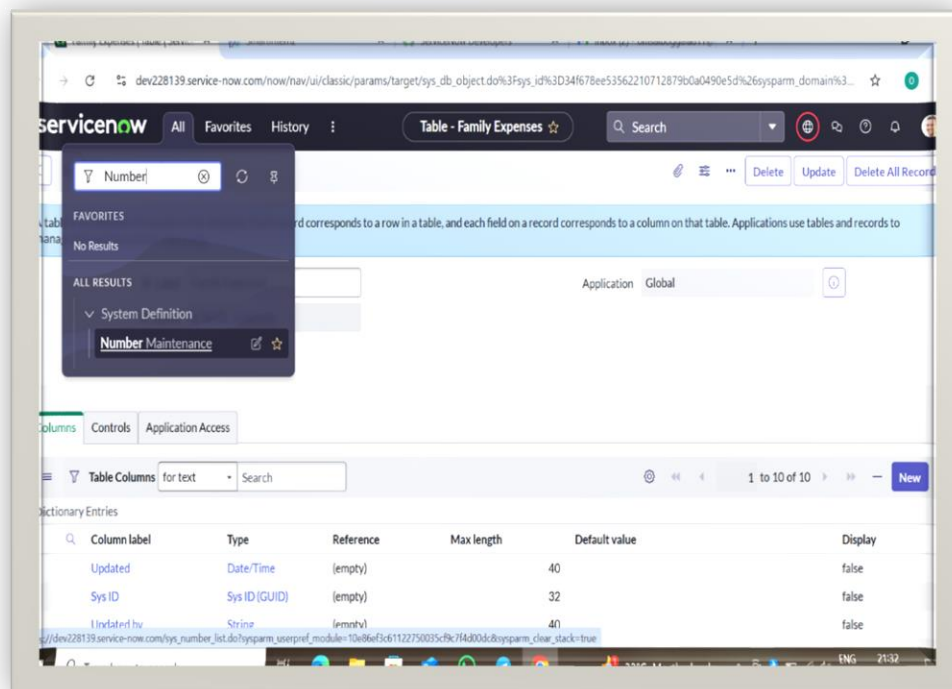
1. Double click on the Number Field/Column.
2. Go down and double click on Advanced view



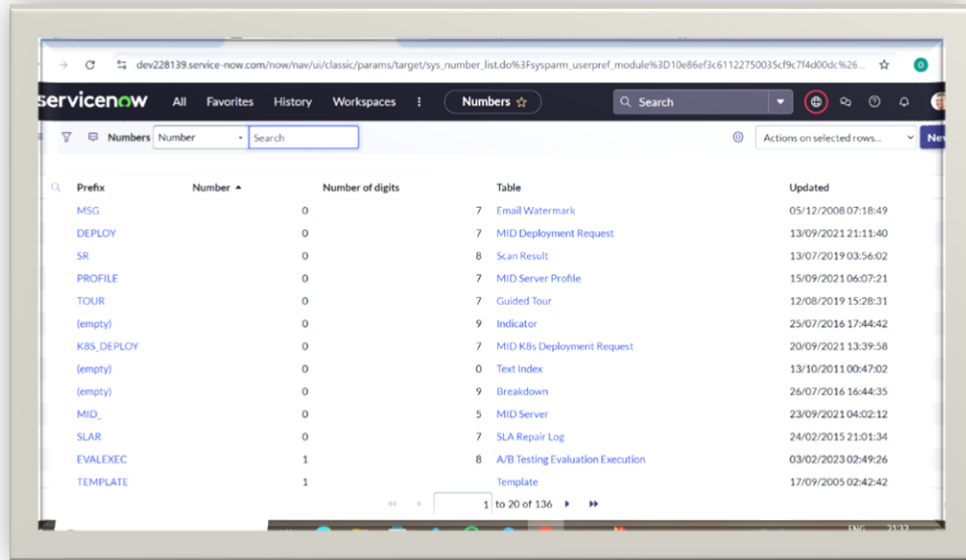
3. In Default Value:
Use dynamic default : check the box
Dynamic default value : Get Next Padded Number
4. Click on Update.



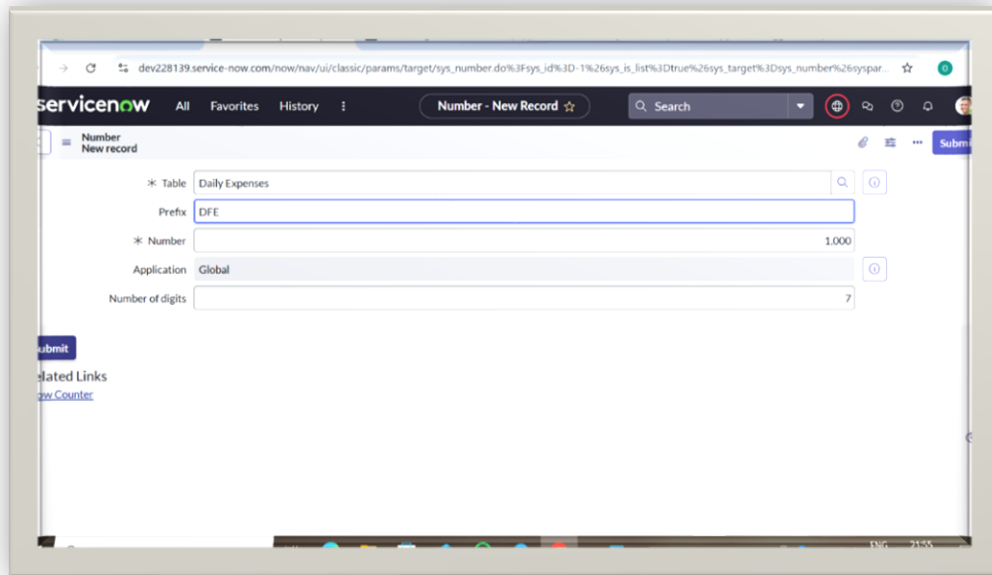
5. Go to All >> In the filter search for Number Maintenance >> select Number Maintenance



6. Click on New.



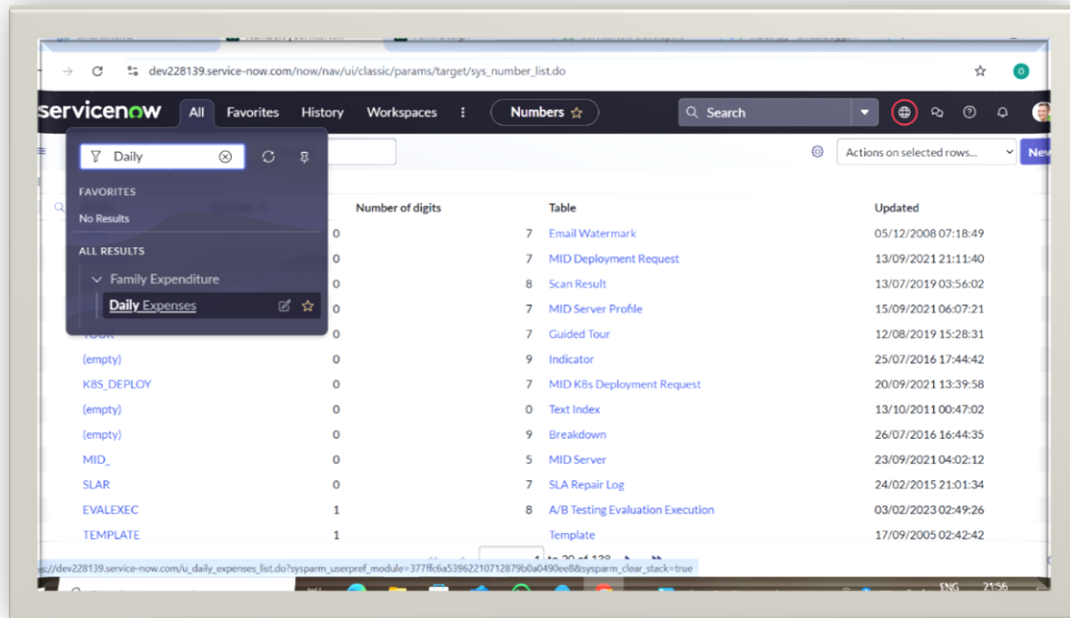
7. Enter the below Details:
Table : Family Expenses
Prefix : MFE



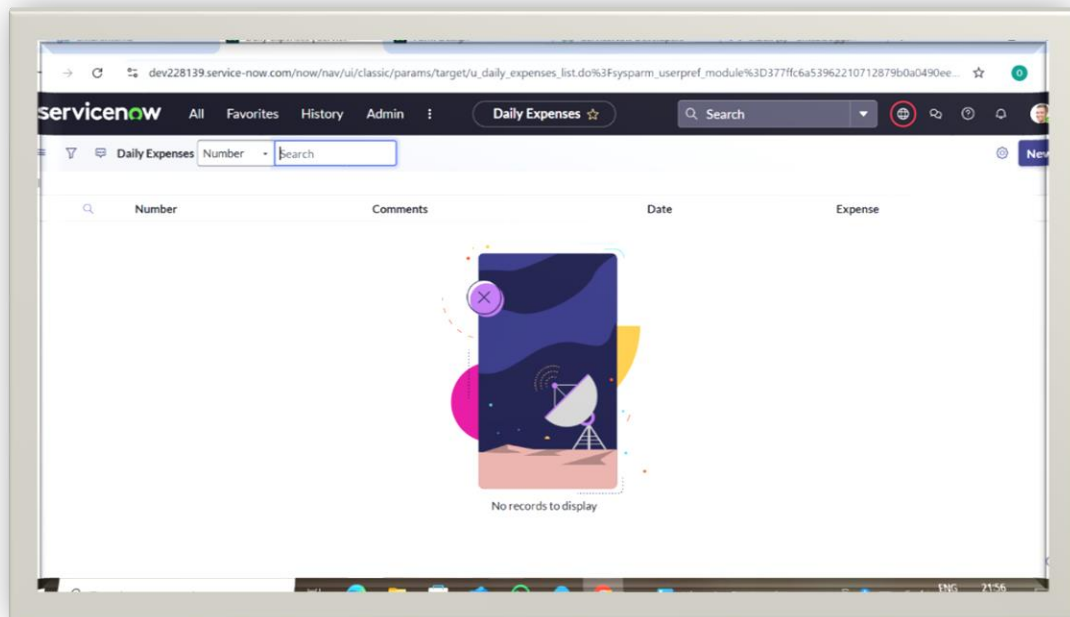
8. Click on Submit.

ACTIVITY-4: Configure the Form

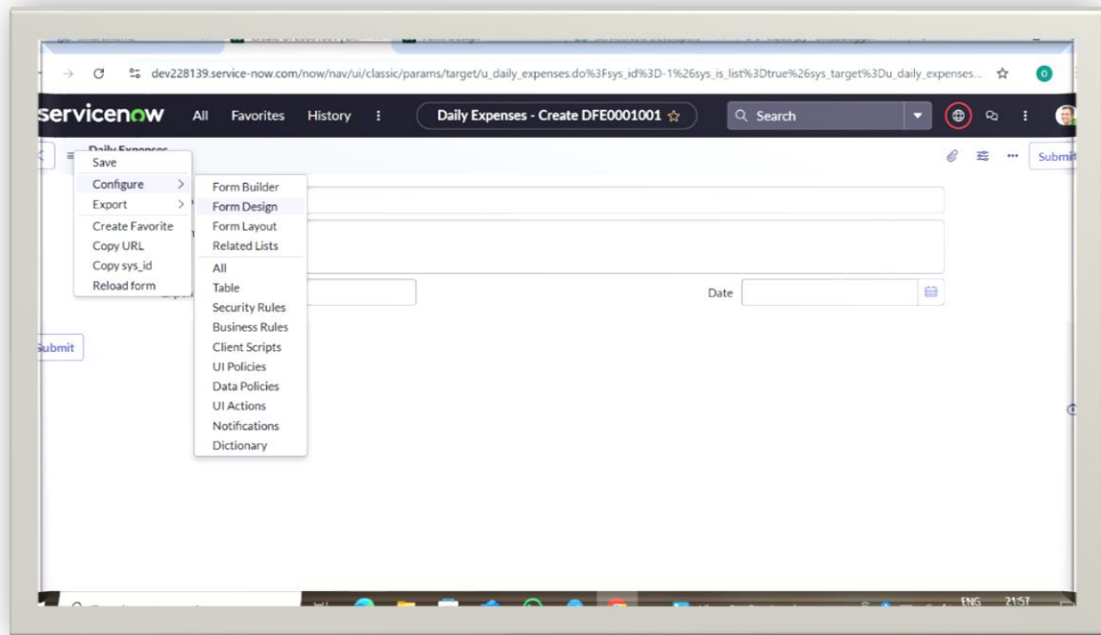
1. Go to All >> In the filter search for Daily Expenses >> Open Daily Expenses



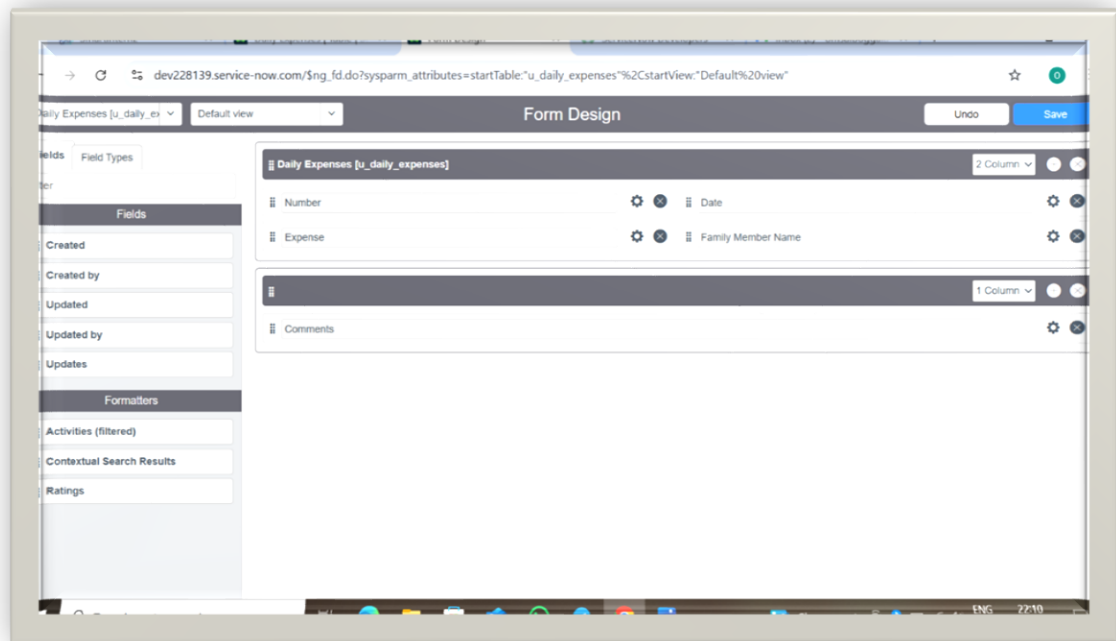
2. Click on New



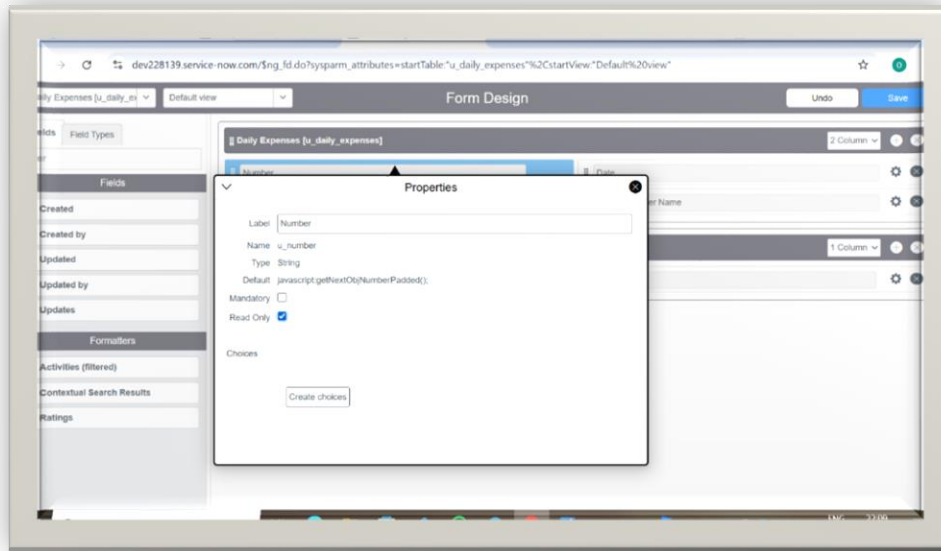
3. Go to the Header and right click there>> click on Configure >> Select Form Design



4. Customize or Drag Drop the form as per your requirement.



5. Make Number Read-Only Field by clicking on the gear icon and checking Read-Only



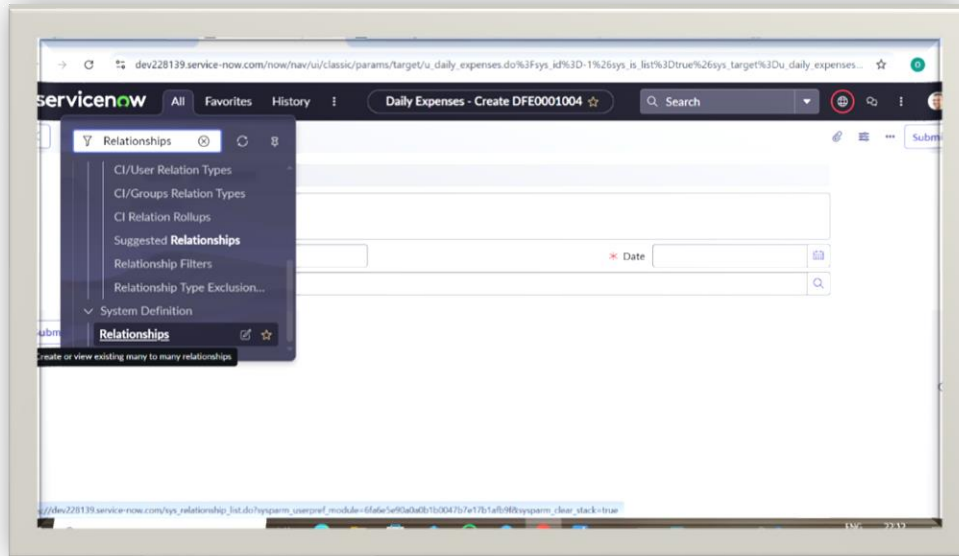
6. Make Date, Family Member Name Mandatory Field by clicking on the gear icon and checking Mandatory
7. Click on Save.

MILESTONE-5:

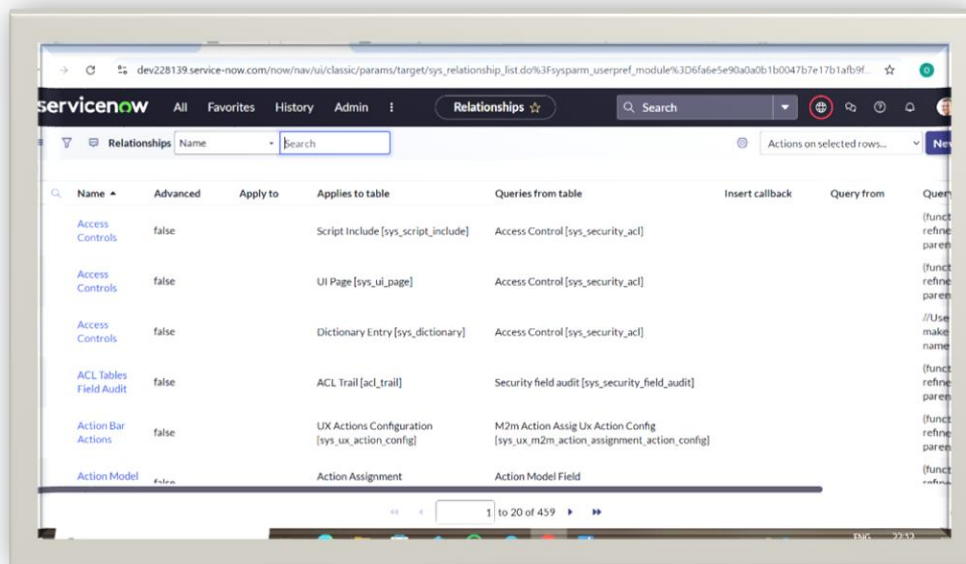
Creation of Relationship

ACTIVITY-1: Creation of Relationship between Family Expenses and Daily Expenses tables

1. Go to All >> In the filter search for Relationships >> Open Relationships



2. Click on New.



3. Enter the details:

Name : Daily Expenses

Applies to table : Select Family Expenses

Queries Fromtable : Select Daily Expenses

dev228139.service-now.com/now/nav/ui/classic/params/target/sys_relationship.do%3Fsys_id%3D-1%26sys_is_list%3Dtrue%26sys_target%3Dsys_relationship%2...

Relationship - New Record

Name: Daily Expenses

Application: Global

Advanced: ☐

Applies to table: Family Expenses [u_family_expense]

Queries from table: Daily Expenses [u_daily_expense]

This script refines the query in current that will populate the related list. For more information about it, its parameters and control variables, see the documentation. See also the article about the recommended form of the script.

Query with ☐ Turn on ECMAScript 2021 (ES12) mode

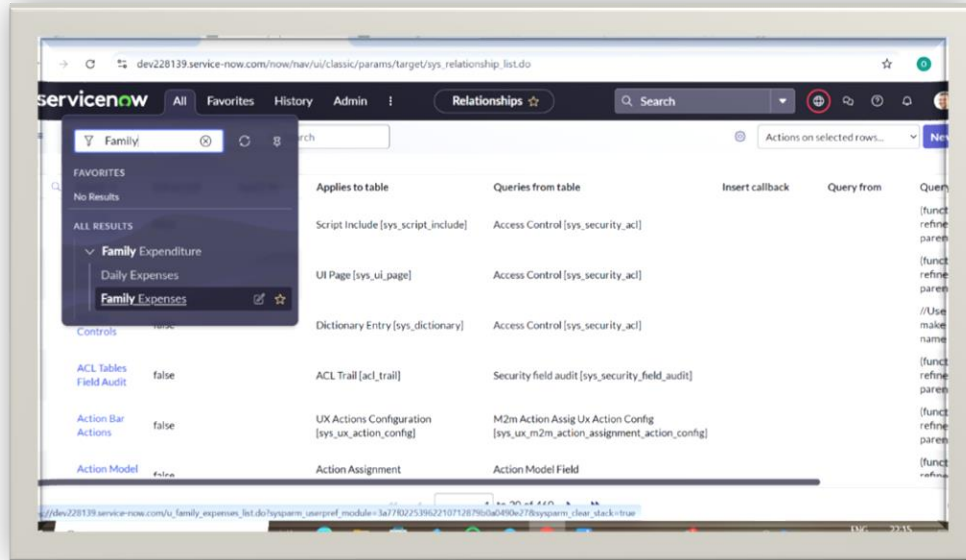
```
1 (function refineQuery(current, parent) {
2
3   // Add your code here, such as current.addQuery(field, value);
4
5 })(current, parent);
```

4. Click Save.

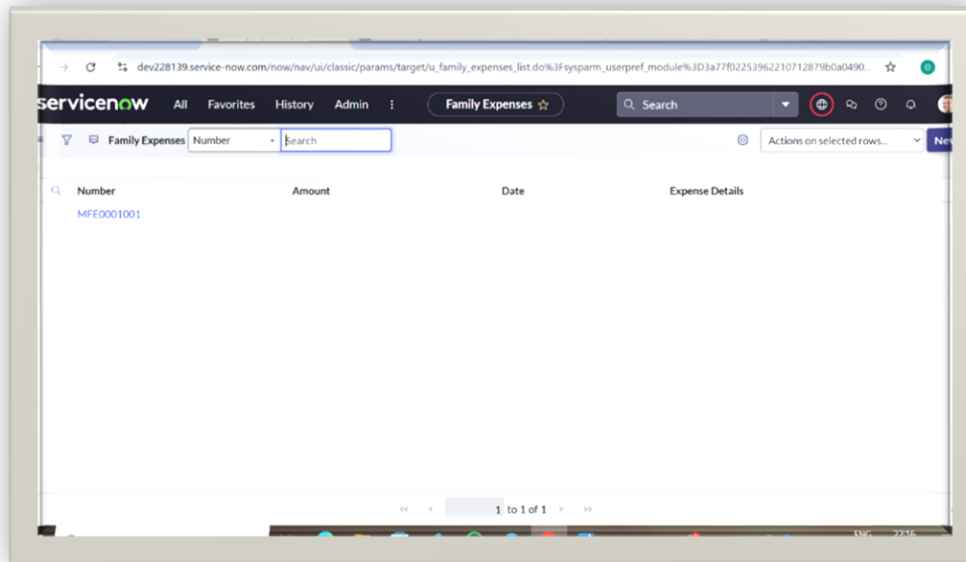
MILESTONE-6:

ACTIVITY-1: Configuring Related List on Family Expenses

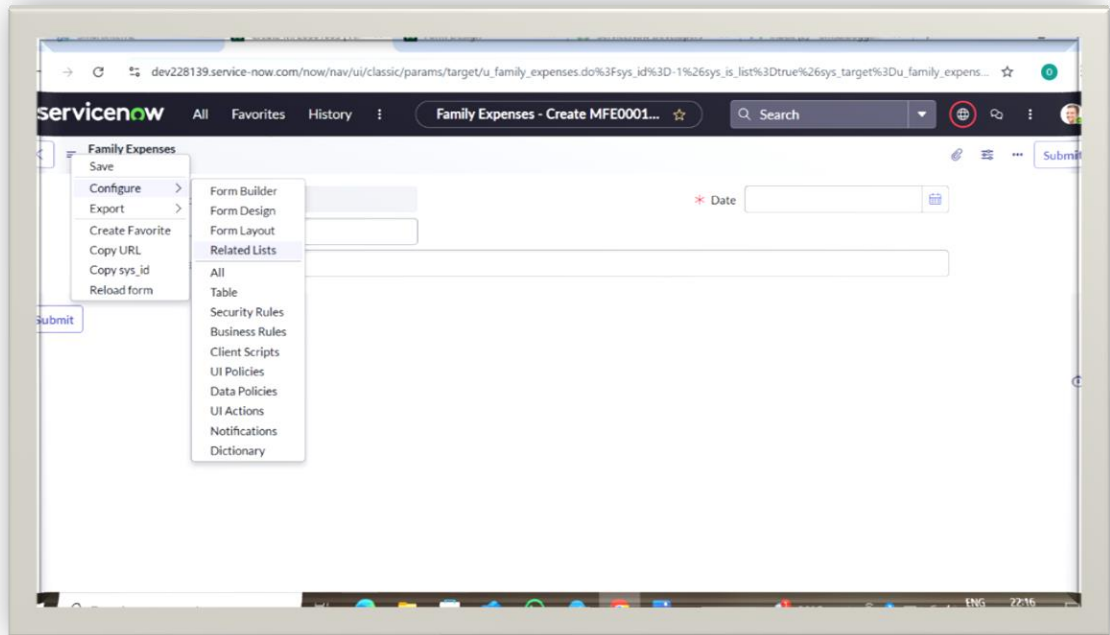
1. Go to All >> In the filter search for Family Expenses >> Open Family Expenses



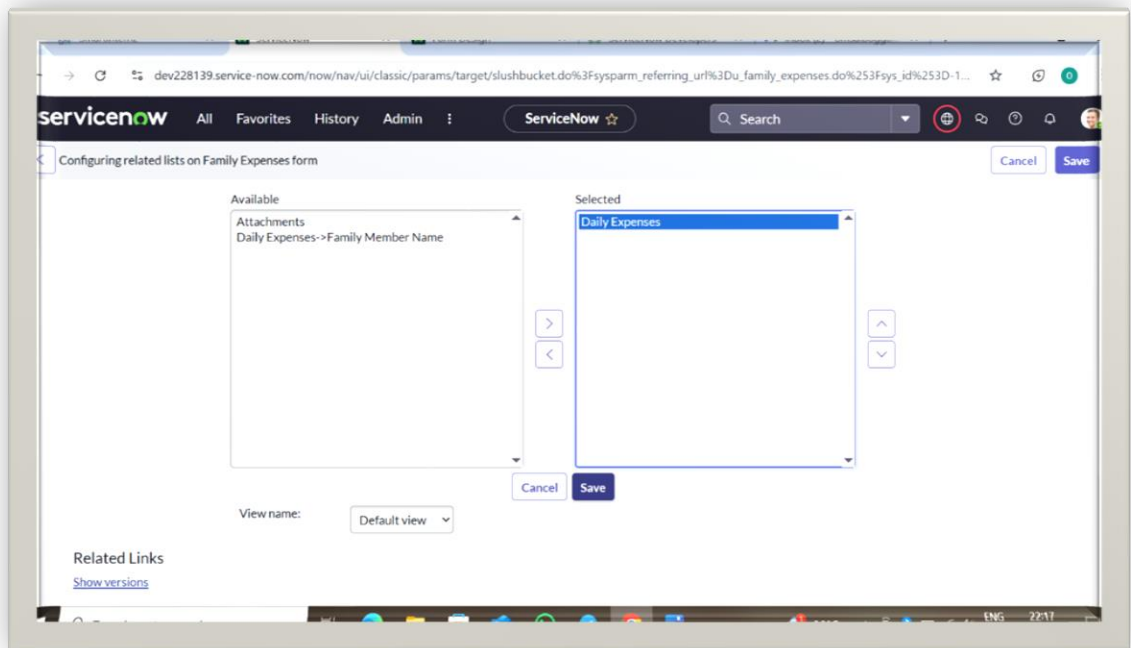
2. Click on New



3. Go to the Header and right click there>> click on Configure >> Select Related Lists



4. Add Daily Expenses to the Selected Area.



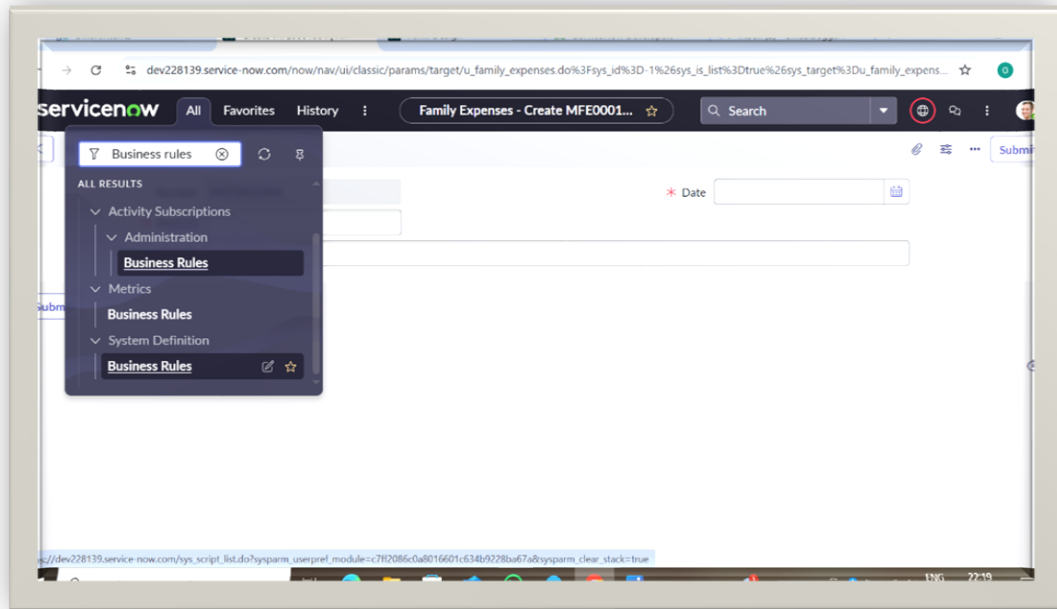
5. Click on Save

MILESTONE-7:

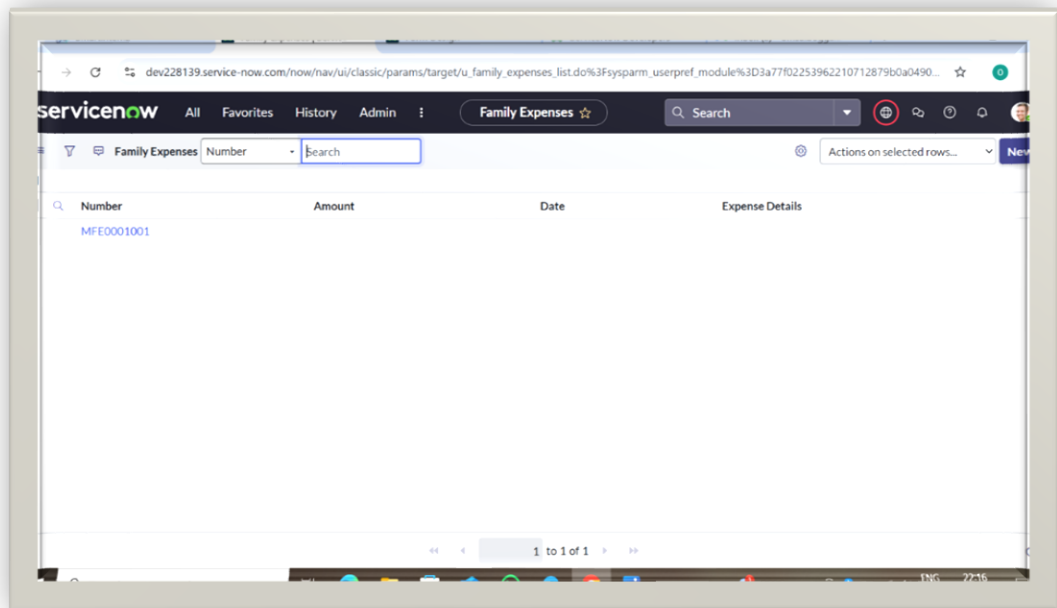
CREATION OF BUSINESS RULES

ACTIVITY-1: Creation of Business Rules

1. Go to All >> In the filter search for Business Rules.

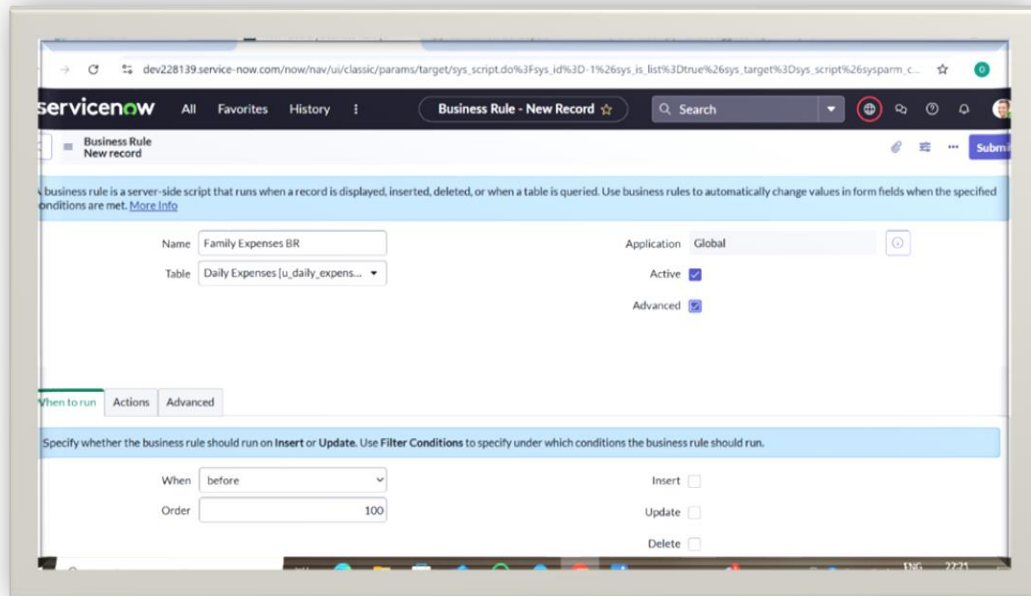


2. Under System Definition Select Business Rules then click on New.



3. Enter the Details:

Name : Family Expenses BR
Table : Select Daily Expenses
Check Advanced



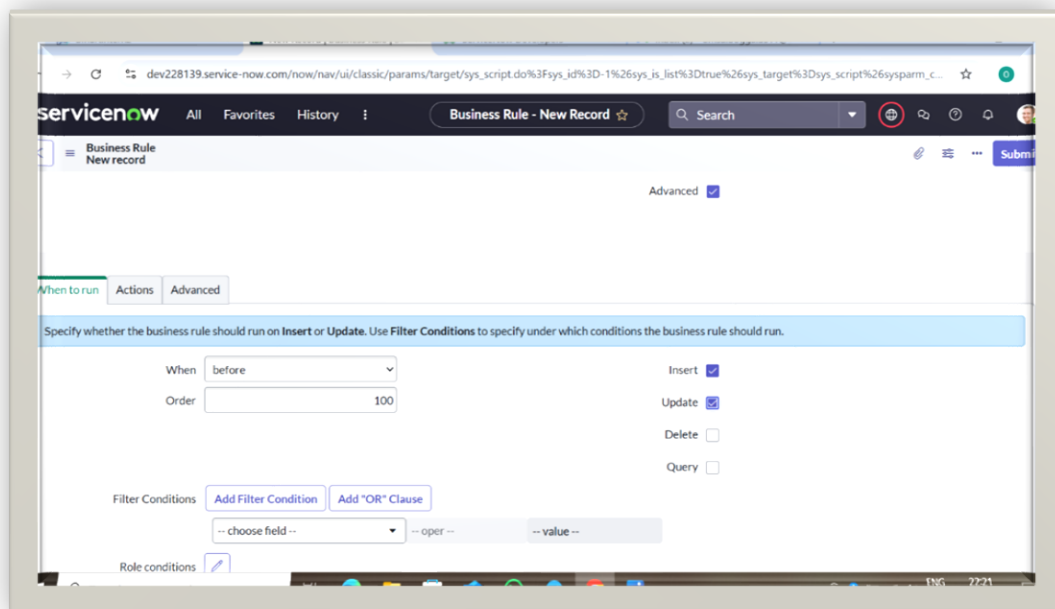
The screenshot shows the 'Business Rule - New Record' form in ServiceNow. The form is titled 'Business Rule - New Record' and includes a search bar and a 'Submit' button. The form fields are as follows:

- Name: Family Expenses BR
- Table: Daily Expenses [u_daily_expens...]
- Application: Global
- Active: ☒
- Advanced: ☒

Below the form fields, there is a section titled 'When to run' with tabs for 'When to run', 'Actions', and 'Advanced'. The 'When to run' tab is selected, and it contains the following options:

- When: before
- Order: 100
- Insert: ☐
- Update: ☐
- Delete: ☐

4. In when to run Check Insert and Update



The screenshot shows the 'Business Rule - New Record' form in ServiceNow, with the 'Advanced' tab selected. The form fields are as follows:

- Name: Family Expenses BR
- Table: Daily Expenses [u_daily_expens...]
- Application: Global
- Active: ☒
- Advanced: ☒

Below the form fields, there is a section titled 'When to run' with tabs for 'When to run', 'Actions', and 'Advanced'. The 'Advanced' tab is selected, and it contains the following options:

- When: before
- Order: 100
- Insert: ☒
- Update: ☒
- Delete: ☐
- Query: ☐

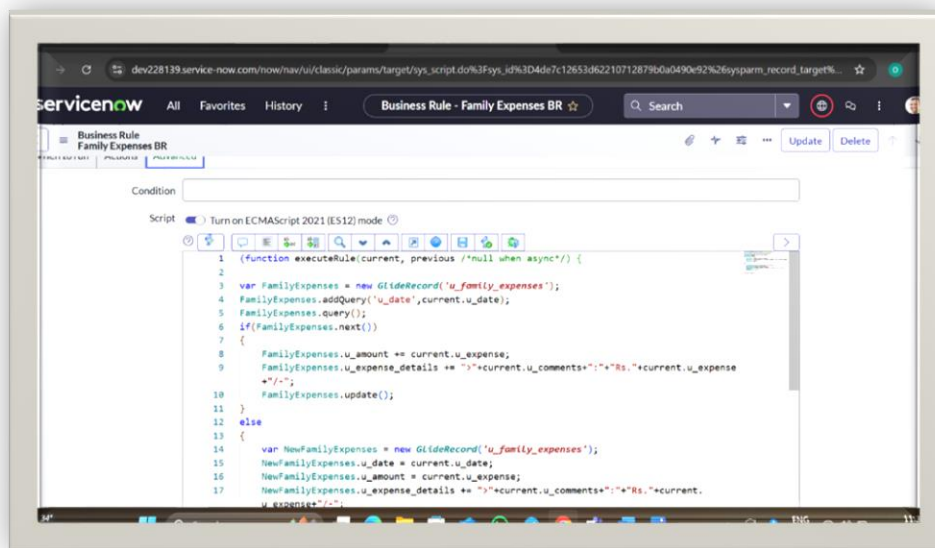
Below the 'When to run' section, there is a section titled 'Filter Conditions' with buttons for 'Add Filter Condition' and 'Add "OR" Clause'. The 'Filter Conditions' section contains the following options:

- Filter Conditions: -- choose field -- -- oper -- -- value --

5. In Advance(we write the code): Write the below code >>

```
(function executeRule(current, previous /*null when async*/)
{
    var FamilyExpenses = new GlideRecord('u_family_expenses');
    FamilyExpenses.addQuery('u_date',current.u_date);
    FamilyExpenses.query();
    if(FamilyExpenses.next())
    {
        FamilyExpenses.u_amount += current.u_expense;
        FamilyExpenses.u_expense_details +=
        ">"+current.u_comments+"":"+Rs."+current.u_expense+"/-";
        FamilyExpenses.update();
    }
    else
    {
        var NewFamilyExpenses = new
        GlideRecord('u_family_expenses');
        NewFamilyExpenses.u_date = current.u_date;
        NewFamilyExpenses.u_amount = current.u_expense;
        NewFamilyExpenses.u_expense_details +=
        ">"+current.u_comments+"":"+Rs."+current.u_expense+"/-";
        NewFamilyExpenses.insert();
    }
}

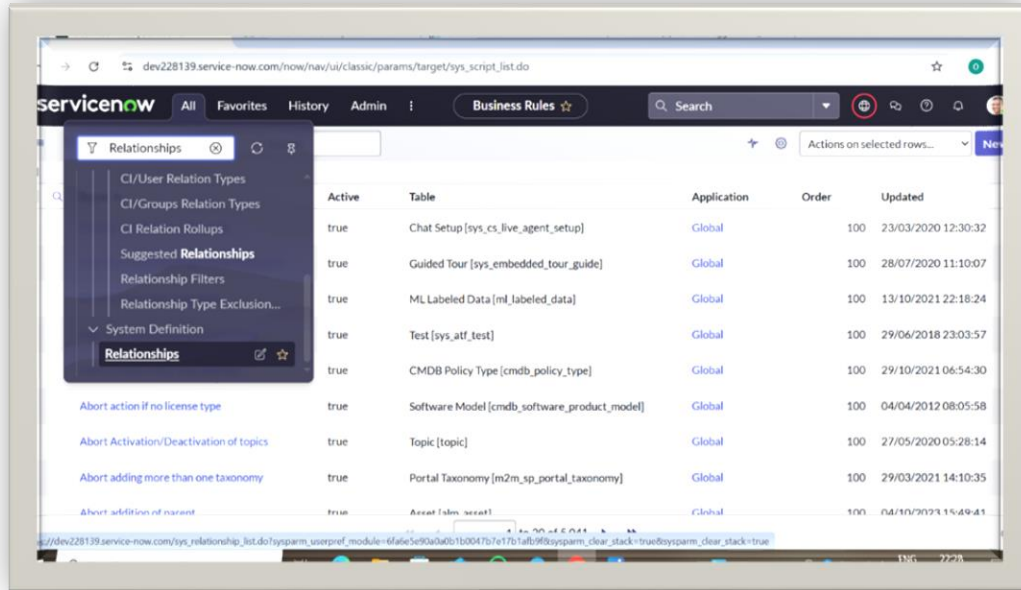
})(current, previous);
```



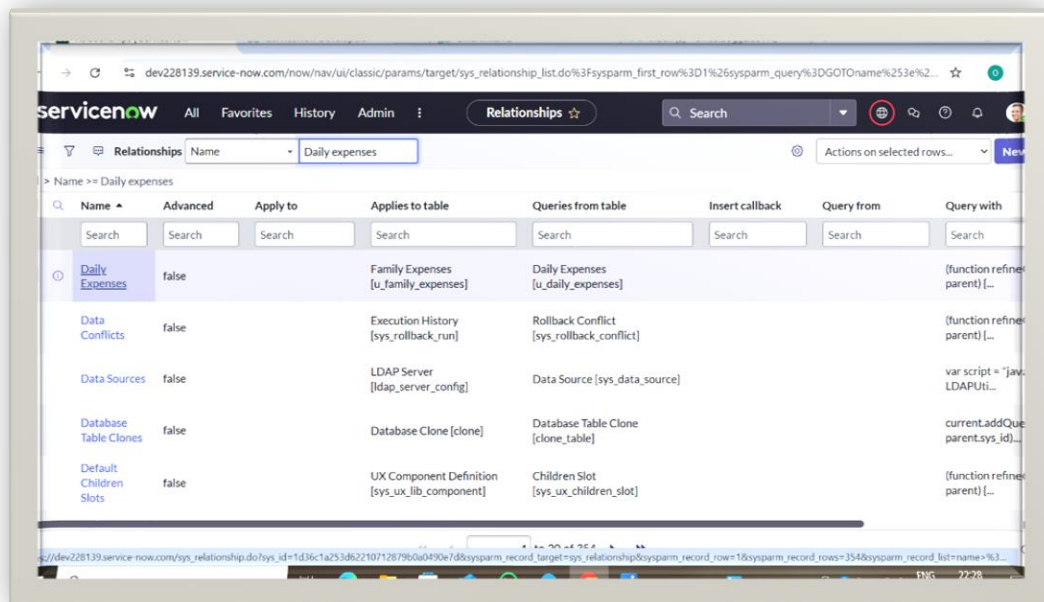
6. Go to the Header and right click there>> click on Save.

MILESTONE-8: Configure the Relationship

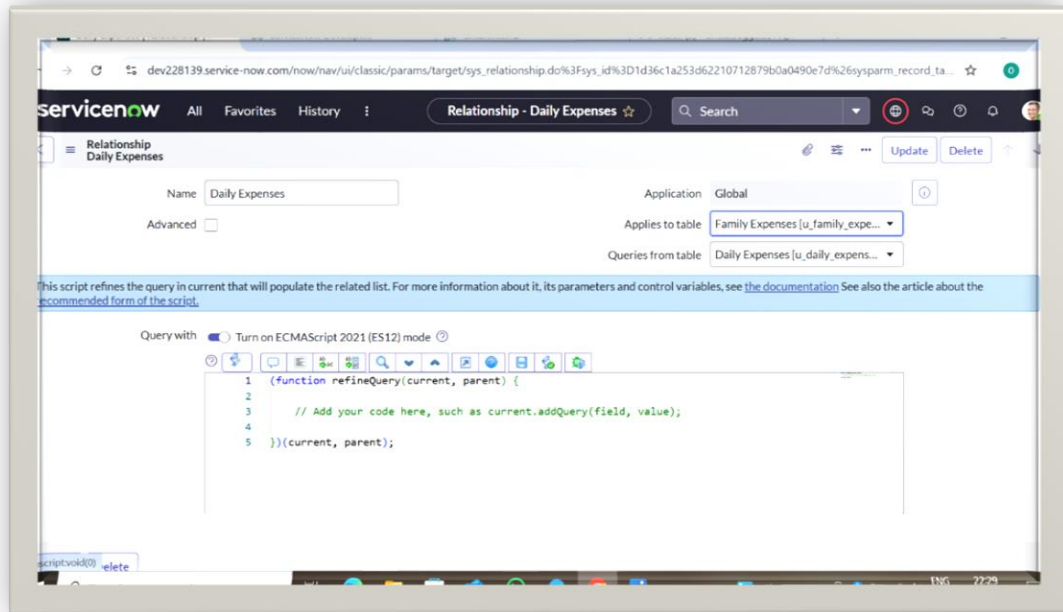
1. Go to All >> In the filter search for Relationships >> Open Relationships.



2. In that, open Daily Expenses Relationship.



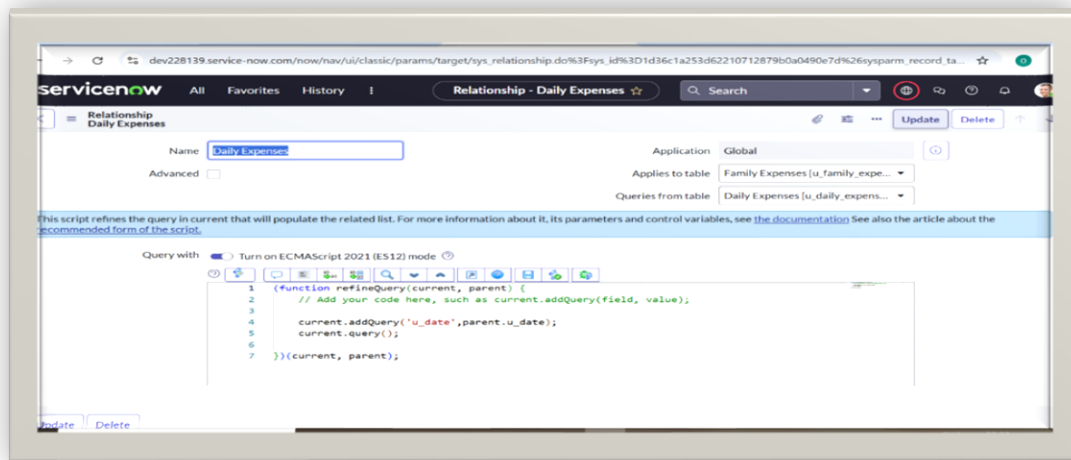
3. For Applies to table : Select Family Expenses.



4. In Query with : write the below Query.

```
(function refineQuery(current, parent) {
```

```
// Add your code here, such as current.addQuery(field, value);  
current.addQuery('u_date',parent.u_date);  
current.query();})(current, parent);
```



5. Click on Update.

MILESTONE-9:

Conclusion

In conclusion, implementing the "Calculation of Family Expenses Using ServiceNow" project offers numerous benefits to the household. By leveraging the robust capabilities of ServiceNow, family members can efficiently submit, track, and manage their expenses in a centralized and streamlined manner.

ADVANTAGES & DISADVANTAGES

Advantages

- **Centralized Expense Management**

All family expense records are stored in one place for easy access and tracking.

- **Automation with Workflows**

Alerts and reminders are automated using ServiceNow Flow Designer.

- **Real-Time Dashboards**

Visual charts allow users to track spending habits and budget usage in real-time.

- **Role-Based Access Control**

Secure separation between Admin and Family Member functionalities.

- **Scalable Platform**

ServiceNow allows future integration (e.g., bank APIs, Excel exports).

- **Mobile Accessibility**

Portal is responsive and usable on mobile devices.

- **Report Generation**

Automated monthly or category-based spending reports help in budgeting decisions.

✖ Disadvantages

- **Platform Licensing Cost**

ServiceNow may require a paid license, which may not be ideal for all families.

- **Overhead for Simple Use Cases**

May be more complex than needed for families with basic expense tracking needs.

- **Learning Curve**

Non-technical users might take time to get comfortable with the portal.

- **Customization Effort**

Initial setup of tables, forms, and workflows takes time and expertise.

- **Limited Offline Access**

Requires internet access; not suitable for offline logging.

✔ Conclusion

The "**Calculating Family Expenses Using ServiceNow**" project successfully demonstrates how a powerful enterprise platform like ServiceNow can be adapted for household-level financial management. By leveraging features such as form automation, role-based access, real-time dashboards, and notification workflows, the system offers families a structured and transparent way to monitor daily spending and stick to budgets.

The application not only improves financial awareness within the household but also fosters better collaboration and accountability between family members. Admins can set limits, track trends, and receive alerts, while members can contribute entries in a user-friendly environment.

Though originally built for enterprise use, ServiceNow has proven flexible and scalable enough for this personal finance use case, opening the door for more creative, non-traditional applications of the platform.



Future Scope

The “Calculating Family Expenses Using ServiceNow” project has strong potential for expansion and improvement. The following are key areas where the solution can grow:

1. Integration with Banking APIs

- Automate expense entry by syncing with bank accounts or digital wallets.
- Real-time import of transactions to minimize manual entry.

2. Mobile App Interface

- Develop a native or hybrid mobile app for quick expense logging and dashboard access.
- Enable offline data entry with sync when online.

3. AI-Powered Insights

- Use AI/ML models to predict future spending trends.
- Provide suggestions to reduce unnecessary expenses or optimize budgets.

4. Voice Assistant Integration

- Add support for Alexa, Siri, or Google Assistant for voice-activated expense logging.

5. Multi-Family or Group Support

- Expand functionality to allow shared dashboards for large joint families or roommate households.
- Add admin-level budgeting for sub-groups.

6. Gamification of Savings

- Introduce point systems or visual achievements for staying under budget.
- Encourage saving through interactive progress tracking.

7. Data Export & Cloud Sync

- Export reports to formats like Excel, PDF, or Google Sheets.
- Sync with cloud storage for data backup and portability.

8. Advanced Security Features

- Enable multi-factor authentication and activity logging.

- Support for audit trails and compliance tracking.

9. ServiceNow Marketplace Publishing

- Convert the solution into a reusable Scoped App that can be shared or sold in the ServiceNow Store.

GITHUB & PROJECT DEMO LINK

<https://github.com/omsai9391/calculating-family-expenses-using-service-now>

**THANK
YOU**